Final Project Part 4

Tyler Asmussen, Odon Ineza, Sarah Senn, and Ava Towery
AME 34560
Virtuous Engineering Design
Professor Goehler

July 24, 2026

Contents

1 List of Potential Customers and Associated Projects

Five engineering concepts for Ireland are compared below. Each subsection lists the idea and the main customers or partners who would buy, fund, or operate it.

1.1 Bus tracking screens with multilingual display

Real-time arrival boards at stops let riders see routes and ETAs without a phone. A multilingual screen helps tourists and non-English speakers. Ireland's National Transport Authority reports that reliability and punctuality on contracted bus services in urban areas remained problematic in 2024, driven in part by staffing shortages and traffic congestion [1]. Clear stop-level information can reduce uncertainty while waiting when schedules slip.
Customers: Dublin Bus, the NTA, and city councils that manage stops.

1.2 Rural transport access

Many rural areas lack reliable buses. Residents without cars—especially older adults and students—often cannot reach jobs, clinics, or school on their own.
Customers: county councils, rural NGOs, Ireland's Department of Transport, and EU rural-development programs.

1.3 Water-based energy generation

Ireland imports most of its energy. Hydro, tidal, and wave systems could add domestic renewable supply using rivers and coastlines.
Customers: National government, the Electricity Supply Board, private developers, and EU renewable-energy programs.

1.4 Corner collision sensors for rural roads

Narrow country roads with blind bends create head-on near-misses. Sensors at a corner could detect traffic on one leg and flash a warning to drivers on the other.
Customers: Transport Infrastructure Ireland, local councils, road-safety groups, and insurers interested in fewer crashes.

1.5 Drain blockage sensor to prevent flooding and standing water

Since it is known to rain heavily in Ireland and Dublin is on the coast, it is subject to flooding. Installing sensors in drains to monitor water levels and identify debris blockages would reduce the probability of flooding early on. Maintenance teams could be alerted immediately and clear drains before standing water occurs.
Customers: Dublin City Council, local authorities, road maintenance agencies, and water management organizations.

2 Research Context

Having selected bus tracking screens with multilingual display for deeper design work, the team consolidated stakeholder needs, legislative floors, and review evidence before ranking customer requirements in Section 3. The product is a trustworthy, accessible, multilingual **presentation layer** on top of the National Transport Authority (NTA) Real Time Passenger Information (RTPI) pipeline [2]. Research showed that the worst RTPI failure is not imprecise GPS alone; it is displaying a timetabled arrival as if it were live when the bus has been cancelled—the "ghost bus" problem discussed in Section 2.3.

2.1 Stakeholders

Table 1 lists the customers and partners who influence requirements. Riders at the stop are the primary end users. Institutional buyers—the NTA, bus operators, and local councils—control procurement, data feeds, and stop infrastructure.

Table 1: Stakeholder map

Stakeholder	Role	Primary need
Riders at the stop	End user	Know when the bus is coming without a phone; trust the information
Tourists / non-English speakers	End user	Read arrivals in a familiar language
Older adults and low-tech users	End user	Access RTPI without downloading an app
Riders with disabilities	End user	Perceive and understand information (visual, cognitive, language)
National Transport Authority	Buyer / RTPI operator	Unified, accurate RTPI across on-street displays, TFI Live, and third-party feeds
Dublin Bus / Go-Ahead Ireland	Operator	Supply AVL data; meet Schedule 14 RTPI obligations
City / county councils	Infrastructure partner	Maintain stop poles, shelters, power; align with designated stop locations
Department of Transport	Policy funder	Deliver statutory real-time passenger information

2.2 Legislative and Contractual Non-Negotiables

Design choices must respect legal, contractual, and policy floors summarized in Tables 2–4. Tier 1 rules draw on EU passenger rights, Irish accessibility regulations, and NTA contract schedules. [3, 4, 5]. The group consulted legal documents to draw conclusions made in the tables below.

Table 2: Tier 1 legal non-negotiables

Rule	Source	Design impact
Accessible information formats for disabled persons	EU Reg 181/2011	Screen UI must meet accessibility standards, not only visual text
Disruption info within 30 min of scheduled departure	EU Reg 181/2011, Art. 20	Must show cancelled, delayed, diverted—not only countdown ETAs
Irish + English language baseline	S.I. 636/2023	Mandatory languages; additional languages are a plus
Interactive RTPI screens must be accessible	S.I. 636/2023; NTA EAA guidance	WCAG 2.1 AA for procured services
NTA owns stop-level customer information	DBDA Schedule 14	Product must align with NTA, not operator-only deployment
Stops are legally designated locations	Road Traffic Act 2002, s. 16	Stop ID and placement must match official registry (± 3 m)

Table 3: Tier 2 contractual non-negotiables (NTA–operator)

Rule	Source	Target
RTPI includes cancellations, curtailments, diversions	Schedule 14 Annex D	No ghost bus displays
Unplanned disruption in feed	Schedule 14 Annex D	≤ 15 min from control-room notification
ETA availability	Schedule 14 Annex D	$\geq 99\%$ of stop-service intervals
Accurate clear-down on arrival/departure	Schedule 14 Annex D	ETA removes when bus passes stop
Not-in-service buses hidden from RTPI	Schedule 31.15.6	100% filter compliance
Planned disruption on RTPI signs	Schedule 5	≥ 10 business days notice
Single national AVL feed (NGAVL)	NTA Trapeze contract	Consume NTA-standard data

Table 4 summarizes how ghost buses and RTPI policy priorities shape design [6].

Table 4: Tier 3 policy drivers

Issue	Source	Implication
Ghost buses	NTA Parliamentary Q&A	Show CANCELLED vs LIVE vs SCHEDULED states
RTPI is statutory public service	gov.ie RTPI policy	Public-interest information duty
NGAVL rollout 2025–2027	NTA NGAVL program	Design for future feed; prototype on current RTPI

2.3 Review and Observation Evidence

Review research drew on Dublin Bus passenger reviews and the TFI Live app [7, 8]. Users frequently report inaccurate real-time data, including buses that appear in the app but fail to arrive—the ghost bus pattern named in the team's accuracy requirement. Other criticisms include an unintuitive interface and difficulty searching for specific stops or routes. Positive reviews from visitors note that the app is helpful when it works, reinforcing the value of clear stop-level information for tourists.

When an operator fails to log a cancellation in the AVL system, TFI Live and on-street displays can revert to showing a timetabled arrival rather than a cancelled state. The Authority reports that RTPI generally works well but that cancellation communication remains inconsistent across apps and on-street signs.

3 Guiding Virtues

In order to create the customer requirements and engineering specifications, observations of existing bus tracking systems, reviews of public transportation services, and consideration of the needs of both local residents and tourists who use Dublin's bus network. The team identified common struggles passengers face when using apps on their mobile devices. For example, the tracking system on the app Transit relies on passengers on the bus to be on the app in order to give a location of the bus. This is a common complaint for many and gave the idea for many of the requirements. The virtues below were used to rank all the requirements based on their perceived importance to passengers, Dublin Bus, NTA, and city councils.
Throughout the process, several virtues guided the team's decisions when translating customer requirements into formal requirements and then into measurable engineering specifications. These virtues were not assigned arbitrarily; they emerged from group discussions in which each team member reflected on their own experiences using public transit in Dublin, on common complaints found in online reviews of Dublin Bus and the Transit app, and on accessibility gaps observed firsthand at bus stops. For each virtue identified, the team asked two questions: which customer requirement does this virtue demand, and how can that requirement be quantified in an engineering specification? The answers drove both the ranking of requirements in Table 5 and the selection of measurable targets in Table 7, consistent with the legislative floors in Tables 2–4. Together, the following virtues shaped a system design that balances technical performance with the real needs of the community:

1. Community awareness was taken into consideration because acknowledging the problems with the current bus tracking system and creating a solution that the community would find helpful is important to create an effective design. By designing for the community rather than the "average user," the system ensures that those who rely on public transit the most—including individuals without smartphones or those who do not speak English fluently—are not left behind.

2. Reflection is important when looking at past customer satisfaction with current solutions and how customers felt the systems could be improved. By honestly evaluating past customer dissatisfaction, the new requirements focus on building a robust, resilient system that fixes old mistakes rather than repeating them.

3. Respect for the commuter means recognizing that their time is their most valuable, non-renewable resource. Inaccuracies within transit systems display a fundamental disregard for the public's schedules, jobs, and personal lives, which is why the system treats every customer as a valued person whose time is worth protecting.

4. Inclusivity guided decisions regarding multilingual communication and public accessibility for those who speak different languages, are disabled, etc. Additionally, the system is intended to serve all passengers who may not own or have access to a smart phone. This was quantified through language support, accessibility compliance, and availability across multiple locations.

5. Accessibility was considered to ensure all passengers can effectively use the system regardless of physical ability. Features such as large text, clear displays, and audible announcements help make transit information available to individuals with visual impairments and other disabilities.

6. Sustainability influenced the development of a durable and efficient system that minimizes energy usage and operating costs over time. Weather resistant components and low maintenance requirements help extend the lifespan of the system and reduce needs for frequent repairs.

7. Reliability was one of the most important virtues because the users must be able to trust the information provided. This influenced requirements such as accurate bus tracking, frequent updates, and dependable operation so that passengers can confidently plan their journeys.

4 Customer Requirements

Building on the stakeholder map, legislative non-negotiables, and review evidence in Section 2, the team ranked customer requirements for the selected bus tracking screen concept. To produce a real-time multilingual bus tracking system, the following customer requirements listed in Table 5 must be considered. These requirements were identified through observations of existing transit systems, bus transportation reviews, and the needs of tourists and locals utilizing the bus systems in Dublin. The requirements are ranked in the left most column of Table 5 from most important to least important based on their value to Dublin bus passengers.

The team assigned display of multiple bus lines and destinations and screen accessibility at bus stops and key locations across the city a shared ranking of 3rd place because they work together to determine how useful the system will be for passengers. Placing displays at bus stops and other high traffic locations ensures that riders can easily access transit information when they need it. At the same time, displaying multiple routes and destinations allows a wider range of passengers to benefit from the system. Without both features, the display would be much less effective at helping users navigate Dublin's bus network.

Similarly, weather resistance and low maintenance requirements were tied for 11th place because both contribute to the long term reliability of the system. Weather resistant displays are necessary to withstand Dublin's climate, while low maintenance designs help reduce service interruptions and repair costs. Together, these features help ensure that the system remains functional and dependable over time.

Table 5: Customer Requirements and Rankings

Rank	Customer Requirement	Description
1	Accuracy [5, 6, 7, 8]	No “ghost buses”; if the tracking system says a bus is on the way it should be reliable.
2	Accessible for users with disabilities [3, 4]	Large text, audible announcements, and wheelchair accessibility improve usability for visually impaired individuals and wheelchair users.
3	Display of multiple bus lines and destinations [2, 8, 7]	Riders should be able to quickly identify the correct bus and destination on the map.
3	Screens at bus stops and popular points [5, 4]	The display is only helpful if it is available at multiple locations that passengers regularly encounter.
5	Delay and service alerts [3, 5, 7, 8]	Users should be able to understand notifications about cancellations, detours, or major delays.
6	Screen must be easily legible [4, 3, 7, 8]	Wording and images must be readable to the human eye without strain.
7	Frequent updates [5, 2]	Arrival times should refresh often to reflect changing traffic and service conditions.
8	Multilingual understanding [4, 7, 8]	A visual map-based display can be understood regardless of a user’s primary language.
9	No personal device required [5, 4]	Individuals without a mobile device, or whose device has run out of battery, should still be able to determine when and where the bus is.
10	Safeguards [5]	No exposed wires, emergency alerts during active emergencies, hazard-free installation, and user privacy as a priority.
11	Weather resistant [5, 2]	Displays must withstand Ireland’s rain, wind, and varying temperatures; electrical components remain safe in wet conditions.
11	Low maintenance requirements [5]	Transit agencies benefit from systems that require minimal repairs and do not degrade over time.
13	Environmentally friendly [2, 5]	Low-energy devices reduce operating costs and support Dublin’s sustainability goals.
14	Easy integration with existing systems [2, 5]	Minimize implementation costs by integrating with existing transit infrastructure and databases.

5 Engineering Specifications

The customer requirements identified in Table 5 were translated into measurable engineering specifications in Table 7 that can be used to guide the design and evaluation of the multilingual bus tracking system. In Table 7, the Rank column represents the priorities of the engineering team responsible for designing and implementing the system. These rankings reflect technical considerations such as safety, reliability, maintainability, regulatory compliance, and overall system feasibility. The Customer Requirements (Rank) column on the far right provides a reference to the corresponding customer requirement and its original customer-assigned ranking. Including both rankings allows for direct comparison between customer priorities and engineering priorities, illustrating how user needs were balanced with practical design and implementation constraints when developing the final specifications. The seven virtues identified in Section 3 were considered when developing these engineering specifications.

5.1 Using Value Sensitive Design to Develop Ethical Engineering Specifications

To incorporate engineering ethics quantitatively into the design process, the team applied Value Sensitive Design (VSD) methodology to two virtues previously used to develop customer requirements in Part 2, reliability and inclusivity. In Table 6, for each virtue, conceptual, empirical, and technical investigations were conducted to determine whether additional engineering specifications were needed or if existing specifications already reflected those values. The reliability investigation revealed passenger trust in the system heavily relies on the frequency of delays, inaccurate information, or "ghost buses." As a result a new engineering specification requiring the ghost bus error rate to be less than or equal to 5% was added. The inclusivity investigation confirmed the importance of existing engineering specifications related to screen legibility and multilingual support. Stakeholder analysis, accessibility requirements, and observations of the current transit system demonstrated that clear displays and support for multiple languages are essential for serving tourists, locals, individuals with disabilities, and passengers without mobile phone access. Together these investigations illustrate how ethical values can be systematically translated into measurable engineering requirements that improve system performance and user experience.

Table 6: Value Sensitive Design (VSD) investigations conducted for virtues of reliability and inclusivity.

Virtue	Investigation	Description
Reliability	**Conceptual**	Reliability is the ability of the system to provide accurate and trustworthy transit information so passengers can remain informed when making their own travel decisions. Direct stakeholders include bus riders, locals, tourists, individuals with disabilities, and commuters. Indirect stakeholders include Dublin Bus, the National Transport Authority (NTA), and local governments. For the bus tracking screen system, reliability means that arrival times, cancellations, and service disruptions are communicated accurately and promptly so that passengers can plan their journeys accordingly.
	Empirical	Research conducted through reviews of Dublin Bus and TFI Live revealed frequent complaints about inaccurate arrival times, uninformed delays, and "ghost buses," where a bus never arrives. Stakeholder feedback suggests that passengers become frustrated and lose confidence in public transportation when this is a recurring incident.
	Technical	The technology must actively support reliable information delivery. The system should consume data directly from the NTA feed and distinguish between live arrivals, scheduled arrivals, delays, and cancellations. The display software should automatically remove arrival predictions after a bus passes the stop and update immediately when operators report service disruptions.
Inclusivity	**Conceptual**	Inclusivity requires that all passengers have equitable access to transit information regardless of language, age, disability, or smartphone ownership. Direct stakeholders include locals, tourists, non-English speakers, passengers with disabilities, and passengers without access to a mobile device. Indirect stakeholders include transit operators and local governments. In this project, inclusivity means ensuring that no passenger is excluded from accessing essential travel information because of language barriers, disabilities, or technological limitations.
	Empirical	Stakeholder analysis, accessibility legislation, and observations of current transit systems indicate that many users struggle when information is only available through mobile applications. Reviews from visitors frequently mention the importance of understandable transit information, while accessibility regulations require public information systems to be usable by individuals with disabilities. These observations demonstrate a clear need for inclusive design features.
	Technical	The technology should be designed to support a wide range of users. The display system should provide accurate multilingual information and large text. Information should be available at bus stops and popular locations across the city for riders without smartphone access. The interface should be evaluated against accessibility standards to ensure the design does not unintentionally exclude any stakeholder groups.

The VSD investigations in Table 6 reflect several core principles from the NSPE Code of Ethics. Most directly, both the reliability and inclusivity investigations align with the canon requiring engineers to hold paramount the safety, health, and welfare of the public, as the Rank system serves a broad and diverse ridership whose ability to navigate the transit network safely depends on accurate, accessible information. The ghost bus error rate specification (5%) addresses the professional obligation requiring engineers to avoid conduct or practice that deceives the public; displaying incorrect or phantom bus arrivals constitutes a form of misinformation that erodes passenger trust and causes tangible harm. The inclusivity investigation connects to the obligation that engineers strive to serve the public interest—here operationalized through multilingual support and screen legibility requirements that ensure the system remains usable by tourists, non-English speakers, passengers with disabilities, and those without smartphone access. Finally, both investigations reflect the obligation that engineers accept personal responsibility for their professional activities, as the team proactively anticipated and mitigated potential harms through systematic ethical analysis rather than treating ethics as an afterthought. Together, these principles reinforce that incorporating VSD into the engineering design process is not merely a procedural exercise but a professional duty.

5.2 Derivation of Ethical Engineering Specifications

The final set of engineering specifications was not defined arbitrarily; instead, it was derived through a structured translation process linking customer requirements, stakeholder needs, Value Sensitive Design (VSD) virtues, and observed system failures in existing Dublin RTPI systems. In particular, five key engineering specifications were newly defined or significantly refined during this translation process: (1) Ghost Bus Error Rate, (2) GDPR Compliance Coverage, (3) Vandalism/Structural Downtime Resiliency, (4) Hardware Bio-Material Recyclability, and (5) Predictive Routing Algorithmic Fairness Index.

Each of these specifications corresponds directly to one or more ranked customer requirements in Table 5, while also being reinforced by ethical and legal constraints identified in Section 2.

1. Ghost Bus Error Rate was derived primarily from the highest-priority customer requirement, *Accuracy*, which emphasizes that arrival predictions must be trustworthy and free from "ghost buses." Empirical evidence from user reviews and RTPI observations showed that phantom arrivals significantly reduce passenger trust. Through VSD reliability analysis, this issue was formalized into a measurable engineering constraint: the system must maintain a ghost bus error rate of below or equal to 5%. This ensures that canceled or non-existent services are never incorrectly displayed as active arrivals.

2. GDPR Compliance and User Data Privacy emerged from the *Safeguards* customer requirement, combined with legal obligations under EU data protection regulations. Stakeholder analysis identified that the system may process sensitive data such as location history and usage patterns. As a result, a 100% compliance target was established to ensure full adherence to data minimization, consent, encryption, and deletion rights, guaranteeing that no personal data is processed without explicit authorization.

3. Vandalism/Structural Downtime Resiliency was derived from the *Low maintenance requirements* customer requirement and reinforced by sustainability and reliability virtues. Observations of public infrastructure in urban environments highlighted vulnerability to physical damage and environmental wear. This specification ensures that any damaged unit must be repairable within 24 hours of downtime, minimizing disruption to passengers and maintaining continuous service availability.

4. Hardware Component Bio-Material Recyclability was developed from the *Environmentally friendly* customer requirement and the sustainability virtue identified in Section 3. The goal is to reduce life cycle environmental impact by ensuring that at least 60% of hardware materials are recyclable through existing municipal or industrial recycling systems. This directly supports long-term environmental stewardship goals for public transport infrastructure.

5. Predictive Routing Algorithmic Fairness Index was introduced from the *Accuracy* and *Safeguards* customer requirements, expanded through the inclusivity and community awareness virtues. Algorithmic bias in predictive routing systems can disproportionately affect different demographic groups. To mitigate this, a fairness index threshold of at least 60% bias mitigation was defined, ensuring equitable distribution of predictive routing outcomes across user populations.

Together, these five specifications demonstrate how qualitative customer needs and ethical principles are systematically converted into quantifiable engineering targets. This ensures that the final system design is both technically measurable and socially responsible.

Table 7: Engineering specifications for the multilingual bus tracking system

Rank	Metrics	Target	Engineering Units	Customer Requirements (Rank)
1	Safety compliance score (including no exposed wiring, emergency alert functionality, hazard-free installation) [5]	100	% compliance	Safeguards (10)
2	Compliance with NTA Accessibility Standards [4, 3]	≥ 90	Compliance score out of 100	Accessible for users with disabilities (2)
3	Protection Rating [5, 2]	≥ 54	Environmental IP Rating	Weather resistant (11)
4	System integration effort [5, 2]	≤ 80	Engineering hours	Easy integration with existing systems (14)
5	Years until repair [5]	≥ 7	years	Low maintenance requirements (11)
6	Percent error between the predicted time and the time of arrival [5, 7, 8]	≤ 25	% Error	Accuracy (1)
7	Ghost bus [5, 6, 7, 8]	≤ 5	% error rate	Accuracy (1)
8	City-wide screen coverage [2]	≥ 50	% of stops covered	Screens must be accessible at all bus stops and popular points across the city (3)
9	User data privacy and GDPR [4, 5]	100	% of compliance and coverage	Safeguards (10)
10	Average power consumption per display unit [2, 5]	≤ 300	Watts	Environmentally friendly (13)
11	Vandalism/structural downtime resiliency [2, 5]	≤ 24	Hours of maximum downtime	Low maintenance requirements (11)
12	Data refresh rate [5, 2]	≤ 60	Seconds to display	Frequent updates (7)
13	Delay & service alert display [3, 5, 6, 8]	≤ 120	Seconds to display	Delay and service alerts (5)
14	Device free access [5, 2]	100	% of information	Tracking must be accessible by those without personal devices (9)
15	Component hardware bio-material recyclability [2, 5]	≥ 60	% of recyclable material	Environmentally friendly (13)
16	Screen Legibility [4, 3]	$\geq$ 3:1	Contrast ratio	Screen must be easily legible (6)
17	Simultaneously displayed routes [2, 8, 7]	≥ 5	Routes	Display of multiple bus lines and destinations (3)
18	Multilingual Support [4]	≥ 4	Languages	Understandable across multiple languages (8)
19	Predictive routing algorithmic fairness index [7, 8, 6]	≥ 60	% bias mitigation ratio	Accuracy (1)

*- Note: The team ranked the new **bolded** engineering specifications and tied them back to the customer requirements in this table above.*

6 Subsystems and Functions

The finalized engineering specifications developed in Section 5 served as the primary framework for generating the functional concepts presented in this section. Rather than designing a single solution, the team used engineering specifications to identify the key subsystems that the live bus tracking multilingual display system must perform, leading to the creation of subsections such as bus tracking, multilingual display, data accuracy, safety, weather and vandalism resistance, sustainability, and accessibility. Within each subsection, multiple functions were generated to explore different methods of satisfying the associated engineering specifications and customer requirements. Both these subsections and their respective functions are shown in Table 8. This structured approach ensured that every proposed function was directly connected to a measurable design objective.

Several engineering virtues also influenced the brainstorming process and the resulting functional concepts. Community awareness encouraged the team to consider the experiences of local commuters, tourists, and individuals who rely on public transportation without smartphones, leading to concepts that emphasize public accessibility and multilingual communication. Inclusivity and accessibility motivated ideas that accommodate users with disabilities and those who speak different languages through features such as audio announcements, large displays, and language selection options. Reliability remained a central consideration when generating concepts for bus tracking and data validation, ensuring that information displayed to passengers is trustworthy and current. Sustainability influenced concepts involving low-power displays, durable materials, and long-life components that reduce environmental impact and maintenance costs. Respect for passengers' time and daily travel needs motivated the development of functions that prioritize clear communication, accurate arrival predictions, and rapid updates, while reflection on the shortcomings observed in existing display systems inspired alternative approaches designed to avoid common issues such as inaccurate information or poor usability. Collectively, these virtues produced a broad set of functional concepts that balance performance with ethical responsibility and customer needs.

Table 8: Subsystems and their functional concepts

Subsystem	**Technical Approaches / Solutions**
Bus Location Tracking	- Use GPS installed on each bus to transmit location data - Use passenger smartphone locations to estimate bus position - Use RFID tags and roadside readers to detect bus locations - Use cameras and computer vision to identify buses along routes - Use driver check-ins at major stops to update location - Use onboard odometers and route maps to estimate position
Multilingual RTPI Display	- Allow users to select a language with a button - Provide audio announcements in multiple languages - Use symbols and pictograms to reduce language dependence - Cycle through languages automatically on a timer - Display multiple languages simultaneously
Accurate Data Displayed	- Pull information directly from the NTA real-time data feed - Cross-check multiple transit databases before displaying information - Compare GPS data to scheduled route information - Allow drivers to report delays and cancellations directly - Validate bus locations using roadside sensors
System Safety	- Keep all wiring enclosed within locked compartments - Install emergency alert messaging capability - Use shatter-resistant display materials - Include automatic fault detection and shutdown systems - Monitor equipment health remotely and report failures
Environmental Protection & Vandalism Resistance	- Use weather resistant materials - Keep all in protective box - Use material that can be easily cleaned and remains durable - Coat entire piece of equipment in weatherproof coating - Recess the screen inside the enclosure for protection - Use tamper-resistant hardware and locks - Install sensors that detect vandalism or forced entry
Sustainable System	- Use low-energy display technologies - Automatically dim the display when ambient light is low - Use long-life components to reduce replacement frequency
Accessibility and Inclusivity of Design	- Display information using large text - Provide audible announcements of arrival information - Use high-contrast color schemes for visibility - Mount displays at wheelchair-accessible heights - Include braille instructions and labels - Allow users to adjust volume and brightness - Display simplified route maps and icons - Support text-to-speech functionality - Provide information without requiring a smartphone

When choosing between the seven subsystems above, the team was most interested in displaying accurate data and displaying multiple languages. This is primarily because these two concepts follow the original project guidelines most since they focus on accuracy and language inclusivity. The team is most interested in helping passengers which could include a wide variety of individuals, many of which could speak other languages. It was found to be important to the team that everyone is able to understand the bus alerts regardless of the preferred language. For these reasons, the team decided to focus on the subsystem that involves displaying multiple languages on the RTPI.

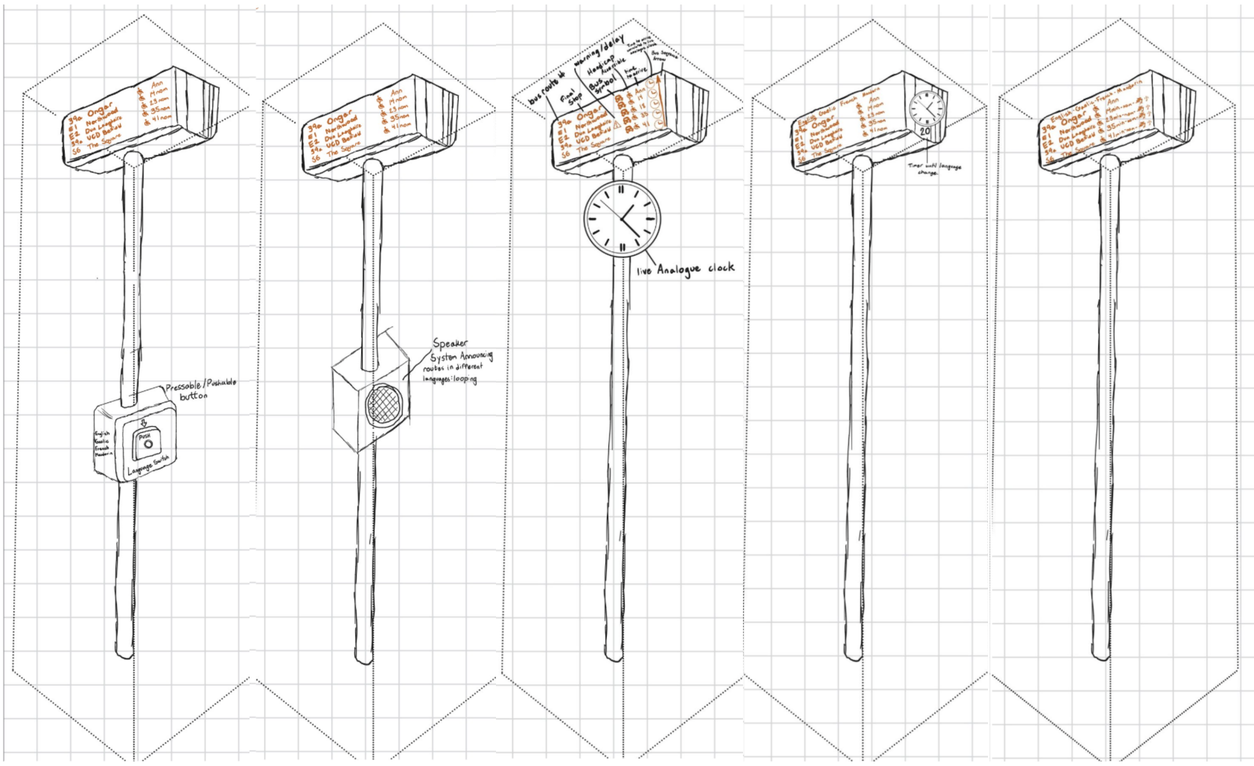
The selected subsystem focuses on how the display communicated transit information in multiple languages to passengers at the bus stops. The purpose is to ensure that arrival times, delays, service alerts, and other essential information can be easily understood by a diverse population including tourists, non English speakers, and locals. By exploring multiple methods of multilingual communication, such as user language selection, audio announcements, symbols and pictograms, automatic language cycling, and simultaneous language displays, the subsystem aims to improve accessibility and inclusivity while maintaining accurate, legible, and device free access to real time bus information.

6.1 Multilingual RTPI Display Function Sketches

Figure 1 illustrates the five functional concepts developed for displaying multiple languages on the realtime bus tracking display. Each sketch represents a different approach to overcoming language barriers while maintaining compliance with the project's engineering specifications for multilingual support, accessibility, and usability. By visualizing these concepts, the team was able to compare alternative methods of communicating transit information to a diverse range of passengers before selecting a final design.

For the multilingual display subsystem, multiple functional concepts were generated to satisfy the engineering specification requiring support for at least four languages while improving usability for tourists, non English speakers, and diverse members of the Dublin community. One concept shown in Figure 1a allows users to manually select their preferred language using a button, enabling personalized interaction and ensuring information is immediately presented in a familiar language. Another concept in Figure 1b provides multilingual audio announcements through integrated speakers, allowing passengers with visual impairments or reading difficulties to receive the same transit information audibly while also assisting users who may not be actively viewing the display. Figure 1c shows a third concept that replaces portions of text with universally recognizable symbols and pictograms, reducing dependence on written language and allowing essential information such as arrival times, delays, accessibility features, and route status to be understood quickly regardless of language proficiency.

The remaining sketches in Figure 1 demonstrate concepts that present multiple languages without requiring user interaction. Figure 1d consists of an approach that automatically cycles through supported languages at fixed time intervals so that different language groups can view the information without touching the display, making it useful in busy public settings. A final concept in Figure 1e displays multiple languages simultaneously on a wider screen, allowing passengers speaking different languages to read the same information at the same time without waiting for a language change. Although this approach requires smaller text and additional display space, it maximizes inclusivity and minimizes interaction requirements. Together, these functional concepts provide multiple feasible methods for satisfying the project's multilingual engineering requirements while promoting accessibility, inclusivity, and ease of use for the broad range of passengers expected to use Dublin's real-time bus tracking system.



(a) Language selection button (b) Audio announcement (c) Symbols and pictograms (d) Language cycle (e) Simultaneous display

Figure 1: Multilingual RTPI display concepts.

All five functional concepts developed for displaying different languages on the live bus tracking display satisfy the project's engineering specifications related to Data Refresh Rate, Delay and Service Alert Display, Device Free Access, Screen Legibility, and Multilingual Support. Regardless of how the language is presented, each concept is intended to display the same continuously updated real-time information received from bus tracking data feed, ensuring that arrival times, delays, and service disruptions are communicated within the required update intervals. Additionally, every concept presents this information directly on the public display, eliminating the need for passengers to rely on a smartphone or personal electronic device. Each design also supports the multilingual engineering specification by communicating transit information in a way that is understandable in multiple languages while striving to maintain clear, legible formatting that can be easily interpreted by users waiting at the stop.

Although all five concepts satisfy these engineering specifications, each does so with different strengths and trade-offs. The button language selection approach provides the highest level of personalization by allowing users to immediately switch to their preferred language, but it requires physical interaction and may create delays when multiple passengers wish to use the display simultaneously. The multilingual audio announcement concept extends accessibility beyond visual communication by assisting passengers with visual impairments or reading difficulties, although it may be less effective in noisy environments and could become disruptive if announcements occur too frequently. The use of symbols and pictograms minimizes language dependence and can communicate common information quickly, but complex service alerts or detailed messages may still require accompanying text to avoid ambiguity.

The automatic language cycling concept ensures that information is available in multiple languages without requiring any user interaction, making it well suited for busy public environments. However, passengers may have to wait for their preferred language to appear, which could reduce convenience when buses are arriving imminently. The simultaneous multilingual display concept eliminates this waiting time by presenting several languages at once, providing immediate access for diverse users and making it particularly effective at satisfying the multilingual support requirement. Its primary drawback is that dividing screen space among multiple languages reduces the amount of space available for each, potentially decreasing text size and overall legibility if not carefully designed. Together, these functional concepts demonstrate multiple viable methods of meeting the project's engineering specifications while balancing accessibility, usability, clarity, and inclusivity in different ways.

6.2 Building Models for the Multilingual RTPI Display

Following the brainstorming work summarized above and illustrated in Figure 1, the team developed quantitative models for the Multilingual RTPI Display subsystem. This step translates the five functional concepts into measurable relationships between design variables (choices the engineer can adjust) and state variables (performance outputs used in the decision matrix). The modeling structure follows the Lecture 19 projectile-motion pattern: each concept has unique design variables, but all concepts are evaluated on the same shared state variables so the alternatives can be compared fairly.

6.2.1 Selected Subsystem

The team selected the Multilingual RTPI Display subsystem for detailed modeling because it directly addresses the project's inclusivity and multilingual customer requirements. This subsystem determines how arrival times, delays, service alerts, and other essential transit information are communicated to passengers at bus stops in multiple languages. The five functional concepts modeled in this section are:

1. Button language selection
2. Multilingual audio announcements
3. Symbols and pictograms
4. Automatic language cycling
5. Simultaneous multilingual display

Each concept offers a different way to satisfy the engineering specifications for multilingual support ($N_{\text{lang}} \geq 4$ languages), screen legibility ($CR_{\text{leg}} \geq 3{:}1$ contrast ratio), and NTA accessibility compliance ($A_{\text{nta}} \geq 90$).

6.2.2 Modeling Approach

All five concepts are evaluated using the same three state variables listed in Table 10. Each concept uses its own unique design variables to determine the values of those shared state variables. Table 9 summarizes how the concepts differ in their design-variable choices while sharing the same performance outputs. The team followed three rules when building these models: (1) pick three state variables that define performance of the multilingual display subsystem; (2) evaluate every brainstorming concept using those same three state variables; and (3) give each concept its own unique design variables that feed into equations for those state variables. Design variables do not enter the decision matrix directly; they are used first to compute state-variable values, which are then normalized to 0–10 scores using `calculateMoM.m`.

Table 9: Design variables and shared state variables for each multilingual display concept.

Concept	**Design Variables (unique to each concept)**	**Shared State Variables**
Button Language Selection	Button size, screen area, menu depth	N_{lang}, CR_{leg}, A_{nta}
Audio Announcements	Audio memory, speaker count, bitrate	N_{lang}, CR_{leg}, A_{nta}
Pictograms	Number of symbols, icon size	N_{lang}, CR_{leg}, A_{nta}
Automatic Cycling	Cycle interval, display duration	N_{lang}, CR_{leg}, A_{nta}
Simultaneous Display	Screen area, font size, layout density	N_{lang}, CR_{leg}, A_{nta}

6.2.3 State Variables

The three state variables in Table 10 define subsystem performance for every concept. They were drawn from the team's engineering specifications and the Measures of Merit framework so that the models remain traceable to ranked customer and engineering requirements. Using a single shared set of state variables ensures that a high score on N_{lang} for one concept means the same thing as a high score for another concept, even when the underlying hardware or interface differs.

Table 10: Shared state variables for the multilingual RTPI display subsystem.

State Variable	**Symbol**	**Definition**	**Engineering Target**
Active Localized Multilingual Support	N_{lang}	Number of languages actively supported at the display	$N_{\text{lang}} \geq 4$
Screen Legibility	CR_{leg}	Contrast ratio between text/symbols and background	$CR_{\text{leg}} \geq 3$
NTA Accessibility Standards Compliance	A_{nta}	Compliance score with NTA accessibility standards (out of 100)	$A_{\text{nta}} \geq 90$

These three variables apply to every functional concept. Concepts without a visual screen (for example, audio-only announcements) still report CR_{leg} as a fixed nominal value so that every concept can be scored through the same normalization function and compared in the decision matrix.

6.2.4 Quantitative Models by Functional Concept

For each concept below, the team defined design variables unique to that interface, wrote governing equations that map those variables to the three shared state variables, and identified how the resulting values will be scored with `calculateMoM.m` using the bounds in Table 16. In each case, the engineer selects feasible combinations of design-variable values, evaluates the labeled equations for that concept, and records the normalized scores for concept selection.

Concept 1: Button Language Selection.

In this concept, the passenger selects a preferred language via a button. The model relates button format choices to language capacity, legibility, and accessibility compliance. Table 11 defines the design variables; Equations (1)–(3) convert them into state-variable values.

Table 11: Design variables for Concept 1 (Button Language Selection).

Symbol	Name	Role
A_b	Button area	Relates to button size; larger buttons reduce how many languages fit on screen
F_s	Font size	Controls text size on the display
A_s	Screen area	Total usable display area

The state variable models for Concept 1 are:

$$N_{\text{lang}} = A_s/A_b \tag{1}$$
$$CR_{\text{leg}} = 0.8 \times F_s \tag{2}$$
$$A_{\text{nta}} = 50 + 0.3 \times A_b + 2 \times F_s \tag{3}$$

Concept 2: Audio Announcements.

This concept delivers spoken transit information without relying on a visual display. Language capacity depends on available audio storage, and accessibility is driven primarily by announcement quality rather than on-screen contrast. Table 12 lists the design variables; Equations (4)–(6) define the state-variable relationships.

Table 12: Design variables for Concept 2 (Audio Announcements).

Symbol	Name	Role
M_a	Audio memory	Total storage available for audio files
S_l	Storage per language	Memory required per supported language
Q_a	Audio quality score	Quality of recorded announcements (0–100)

The state variable models for Concept 2 are:

$$N_{\text{lang}} = M_a/S_l \tag{4}$$
$$CR_{\text{leg}} = 10 \quad \text{(fixed; no screen dependency)} \tag{5}$$
$$A_{\text{nta}} = 70 + 0.3 \times Q_a \tag{6}$$

Concept 3: Pictograms.

Universal symbols reduce dependence on written language by communicating route, time, and alert information through icons. Table 13 defines the vocabulary size and icon dimensions; Equations (7)–(9) estimate how symbol richness and size affect multilingual reach, legibility, and compliance.

Table 13: Design variables for Concept 3 (Pictograms).

Symbol	**Name**	**Role**
N_p	Number of symbols	Count of pictogram symbols in the display vocabulary
I_s	Icon size	Physical size of each symbol

The state variable models for Concept 3 are:

$$N_{\text{lang}} = 1 + 0.5 \times N_p \tag{7}$$
$$CR_{\text{leg}} = 0.8 \times I_s \tag{8}$$
$$A_{\text{nta}} = 85 + 0.75 \times N_p \tag{9}$$

Concept 4: Automatic Cycling.

This concept cycles through supported languages on a fixed timer so passengers do not need to interact with the display. The number of languages a rider can see during a wait depends on cycle length and average dwell time at the stop. Table 14 defines the timing variables; Equations (10)–(12) compute the resulting state variables.

Table 14: Design variables for Concept 4 (Automatic Cycling).

Symbol	**Name**	**Role**
T_c	Cycle interval	Time each language is shown before switching (seconds)
T_w	Display duration / average wait time	Total time a passenger waits at the stop (seconds)

The state variable models for Concept 4 are:

$$N_{\text{lang}} = T_w / T_c \tag{10}$$
$$CR_{\text{leg}} = 8 \quad \text{(fixed; one language shown at a time)} \tag{11}$$
$$A_{\text{nta}} = 95 \quad \text{(fixed; timed cycling meets baseline accessibility)} \tag{12}$$

Concept 5: Simultaneous Display.

Multiple languages appear on screen at the same time, eliminating wait time for a preferred language but reducing the space available per language block. Table 15 lists the layout variables; Equations (13)–(15) capture the trade-off between route density, font size, and language count.

Table 15: Design variables for Concept 5 (Simultaneous Display).

Symbol	**Name**	**Role**
A_s	Screen area	Total usable display area
F_s	Font size	Text size for each language block
N_r	Route count / layout density	Number of routes displayed; higher density reduces space per language

The state variable models for Concept 5 are:

$$N_{\text{lang}} = A_s/(F_s \times N_r) \tag{13}$$

$$CR_{\text{leg}} = 0.6 \times F_s \tag{14}$$

$$A_{\text{nta}} = 74 + F_s \tag{15}$$

6.2.5 Scoring State Variables with calculateMoM.m

After computing state variables from each concept's design-variable models, scores are normalized using the team's MATLAB function `calculateMoM.m`. This function applies the minimum and maximum bounds from Table 16 so that values at or below the engineering minimum score 0, values at or above the maximum score 10, and values in between are linearly interpolated. For all three state variables in this subsystem, the direction is `'higher'` because larger values indicate better multilingual coverage, legibility, or accessibility performance. If the state value is less than or equal to the minimum bound, the score is 0; if it is greater than or equal to the maximum bound, the score is 10; otherwise the score is linearly interpolated between 0 and 10 using the relationship $mom\_score = 10 \times (state\_val - min\_val)/(max\_val - min\_val)$.

Table 16: Normalization bounds for multilingual display subsystem models.

MoM	**State Variable**	**min_val**	**max_val**	**Direction**
M_1 — Languages Supported	N_{lang}	4 languages	8 languages	higher
M_2 — Legibility	CR_{leg}	3	15	higher
M_3 — Accessibility	A_{nta}	90	100	higher

6.2.6 From Design Variables to Decision Matrix Scores

Figure 2 summarizes how the team converts concept-level design choices into comparable 0–10 scores for the decision matrix. The workflow has two linked stages. First, each concept variant is defined by a specific set of design-variable values (for example, FC1a and FC1b for Concept 1). Those values are passed through that concept's governing equations to produce the three shared state variables, N_{lang}, CR_{leg}, and A_{nta}. Design variables never enter the decision matrix directly; they only affect performance through these state-variable outputs.

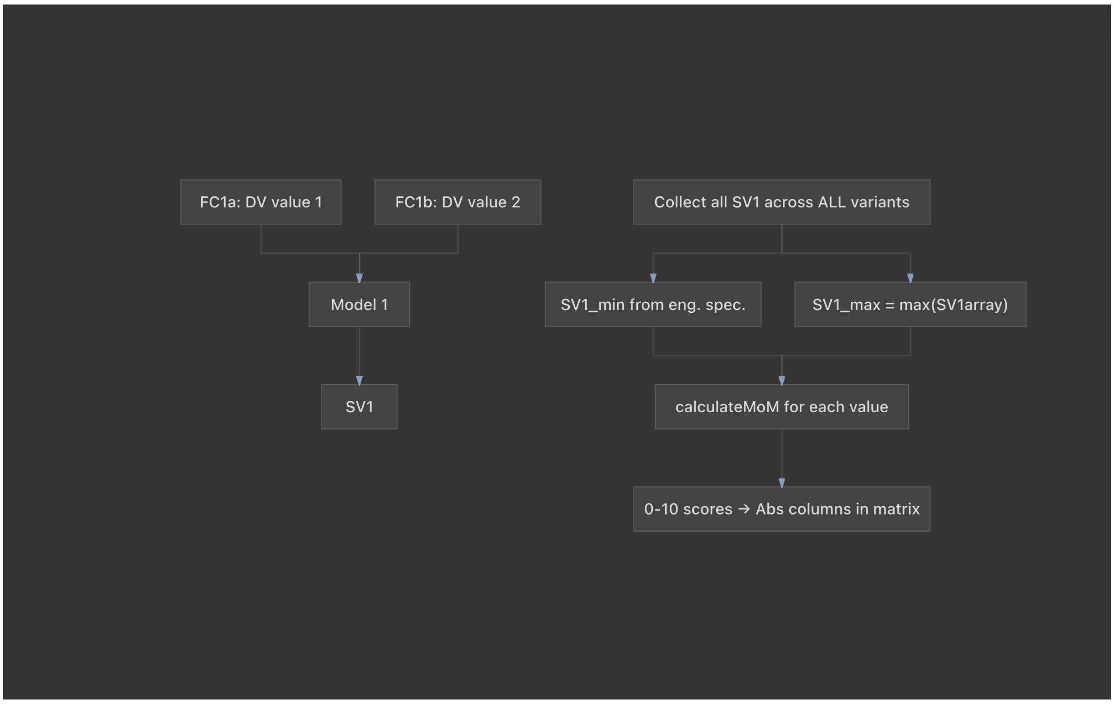


Figure 2: Workflow for computing normalized MoM scores: design variables are mapped to state variables through each concept model, state-variable values are collected across all tested variants, and `calculateMoM.m` converts each value to a 0–10 score for the decision matrix.

Second, scores are normalized across the full tradespace being compared. For each state variable, the team collects every value produced by the MATLAB design sweeps into a single array (for example, $\mathrm{SV1}_{\mathrm{array}}$ for N_{lang}). The lower normalization bound, $\mathrm{SV}_{\mathrm{min}}$, is taken from the engineering specification (Table 16). The upper bound is set dynamically as $\mathrm{SV}_{\mathrm{max}} = \max(\mathrm{SV}_{\mathrm{array}})$ over all concept variants and design combinations tested in that sweep. This ensures that the best-performing variant in the study receives a score of 10 for that criterion, while any variant at or below the engineering minimum receives 0. Each value in the array is then passed to `calculateMoM.m` along with $\mathrm{SV}_{\mathrm{min}}$, $\mathrm{SV}_{\mathrm{max}}$, and the direction of improvement (`'higher'` for all three state variables in this subsystem). The resulting 0–10 scores populate the absolute (Abs.) columns in the decision matrix; weighted (Wgt.) scores are obtained by multiplying each absolute score by the criterion weight.

7 Methods

7.1 Customer Metric Score System

The team scored all five projects on seven metrics. The five metrics originated from an earlier group discussion and virtues: environmental safety, implementation difficulty, accessibility, cost, and utility. Market potential and scalability were later added to meet the assignment's customer-focus requirement. Table 17 defines each metric with a corresponding virtue that it effects, with each metric weighted equally.

Table 17: Evaluation criteria

Metric	Why it matters	Associated Virtues
Environmental safety	For solutions that touch rivers, roads, and countryside; impacts must stay low.	Responsibility, Stewardship
Implementation difficulty	Ideas that cannot reasonably be built or piloted are less suitable for the scope of this course.	Prudence, Practicality
Accessibility	Do the proposals target users without smartphones or steady transport? If so, some people might not access it.	Community Awareness, Respect
Cost	Capital and upkeep must fit real budgets.	Stewardship, Prudence
Utility	This metric emphasizes clear user benefit rather than technological novelty alone.	Service, Compassion
Market potential	There must be identifiable buyers or funders.	Initiative, Prudence
Scalability	A pilot should extend to more sites without a full redesign.	Resourcefulness, Adaptability

Each project's corresponding advantages and disadvantages are ranked based on the metrics it includes. The advantages are ranked by looking at every metric with a positive (+1), a strong positive (+2), or a neutral (0). The disadvantages are similarly ranked with a negative (-1), a strong negative (-2), or a neutral (0). The scores for each project are then summed together to get a total metric score.

7.2 Measures of Merit

To transition from abstract design ethics to quantitative optimization, the team augmented the engineering framework using the parameters derived from the Value Sensitive Design (VSD) tripartite methodology outlined in Section 5.1. This systematic structure links the qualitative customer expectations and ethical values established in Table 6 directly with the mathematical state variables required by the optimization environment.

Each parameter maps to an absolute minimum (x_{min}) or an absolute maximum (x_{max}) boundary depending on whether the goal is to minimize or maximize, using case-sensitive directional strings (`'higher'` or `'lower'`) to translate physical engineering units into normalized utility values from 0 (worst case) to 10 (ideal target).

The integration of the VSD investigations into the optimization framework is structured around the two core virtues:

1. **Reliability Metrics (Ghost Bus Errors):** Based on the conceptual, empirical, and technical VSD findings detailed previously, the new engineering specification for the Ghost Bus Discrepancy Reporting Error Rate (E_{ghost}) was formally integrated. It is embedded into the optimization framework as Row 19 with an operational target threshold of $\leq 5\%$. It is positioned at Rank 2 in the system hierarchy because preserving truthfulness in public data is critical to basic utility.

2. **Inclusivity Metrics (Multilingual & Accessibility Compliance):** The accessibility and multilingual requirements validated during the VSD process were translated into native constraints within the optimization performance loop. These are parameterized as explicit accessibility criteria (A_{nta}) and multilingual capabilities (N_{lang}), ensuring that the interface configuration maximizes demographic reach without compromising core data refresh rates.

By combining these virtue-driven metrics with four additional technical ethics parameters—addressing data justice (GDPR compliance), public asset resiliency (vandalism mitigation), environmental stewardship (recyclability), and algorithmic fairness—the final optimization metrics ensure that systemic equity is embedded directly inside the project's engineering code loops.

The measures of merit in Table 18 capture the engineering parameters that can be optimized and traded against one another during design evaluation. However, four specifications cannot participate in trade-space optimization because they represent absolute legal, contractual, or functional thresholds: any design that fails to meet them is simply non-compliant, regardless of how well it performs elsewhere. These pass/fail requirements are separated into Table 19 to make clear that they are not variables to be minimized or maximized but fixed constraints that every candidate design must satisfy before it can be considered further.

Table 18: Measures of Merit Derived from Value-Sensitive Engineering Specifications

No.	Measure of Merit (MoM)	State Var.	MATLAB Dir.	Abs Min (x_{min})	Abs Max (x_{max})
1	NTA Accessibility Standards Compliance	A_{nta}	'higher'	90%	—
2	Enclosure Environmental Protection Rating	R_{IP}	'higher'	54 (IP54)	—
3	System Integration Development Effort	T_{int}	'lower'	—	80 hrs
4	Asset Operational Lifespan Before Major Repair	L_{maint}	'higher'	7 years	—
5	Real-Time Tracking Prediction Error	E_{predict}	'lower'	—	25%
6	Ghost Bus Discrepancy Reporting Error Rate	E_{ghost}	'lower'	—	5%
7	City-Wide Bus Stop Network Coverage	C_{stop}	'higher'	50%	—
8	Active Display Power Consumption	P_{unit}	'lower'	—	300 W
9*	Vandalism Structural Downtime Resiliency	t_{down}	'lower'	—	24 hrs
10	Live Core Database Refresh Latency	t_{refresh}	'lower'	—	60 s
11	Incident & Emergency Service Alert Latency	t_{alert}	'lower'	—	120 s
12*	Hardware Component Bio-Material Recyclability	M_{recycle}	'higher'	60%	—
13	Screen Legibility Contrast Ratio	CR_{leg}	'higher'	3:1	—
14	Maximum Simultaneous Route Slots	N_{route}	'higher'	5 routes	—
15*	Predictive Routing Algorithmic Fairness Index	I_{trans}	'higher'	60%	—

Table 19: Non-Negotiable Pass/Fail Measures

No.	Requirement	State Var.	Required Value	Justification
1	Safety Compliance Score	S_{comp}	100%	Legal requirement; any exposed wiring or hazard renders the system non-deployable
2	User Data Privacy & GDPR Compliance	P_{gdpr}	100%	Legal requirement under EU data protection law; cannot be below 100%
3	Active Localized Multilingual Support	N_{lang}	≥ 4 languages	Functional requirement; fewer than 4 languages fails to pass the threshold the team set for language inclusivity
4	Device-Independent Information Availability	I_{freed}	100%	Functional requirement; any information gap excludes passengers without smartphones

The measures of merit in Table 18 and the pass/fail measures in Table 19 are intentionally separated because they serve different roles in the design process. The measures of merit are optimizable parameters: values that can be traded, improved, or balanced against one another when comparing design alternatives. The pass/fail requirements, by contrast, are non-negotiable thresholds derived from legal obligations, core functional mandates, and the team's expectations; a design that fails any one of them cannot be considered viable regardless of how well it performs on the remaining metrics. Combining the two in a single table would obscure this distinction and risk treating fixed constraints as if they were adjustable targets.

7.3 Measures of Merit and Linear Normalization Schema

Table 18 codifies nineteen distinct measures of merit derived from the finalized specifications. Each parameter maps an engineering state variable to a standardized, dimensionless 0–10 utility scale. The algorithmic transformation is executed via the MATLAB utility `calculateMoM` (Appendix A), which applies bounded linear interpolation across the domain where $x_{\text{max}} \neq x_{\text{min}}$:

$$U(x) = \max\left(0, \min\left(10, 10 \cdot \frac{x - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}\right)\right) \tag{16}$$

For inverse performance variables designated as `'lower'` (e.g., latency, error rates), the mapping inverts such that smaller physical values yield higher utility scores:

$$U(x) = \max\left(0, \min\left(10, 10 \cdot \frac{x_{\text{max}} - x}{x_{\text{max}} - x_{\text{min}}}\right)\right) \tag{17}$$

Core ranges establish strict engineering boundaries: E_{ghost} spans 0%–5%, legibility contrast ratios (CR_{leg}) span 3:1–15:1, and refresh latencies span 1–60 s. While ethical MoMs utilize broad bounds to accommodate early-stage prototype performance, their presence explicitly constrains tradespace optimization.

8 Results

This section reports the group evaluation outcomes. The team first present virtue-linked advantage and disadvantage tables for all five concepts from Section 1. Then the team gave the numeric score matrix and the bar chart used to compare totals. Together, these results support the discussion and recommendation in Section 6.

8.1 Pro-Con Analysis Tables with Virtues

Tables 20–24 summarize trade-offs. Each row lists an advantage or disadvantage and its respective score. Each advantage and disadvantage is given a series of points based on the metrics in Table 17. The number in the Metric Score section of Tables 20–24 is the sum of the scores for each advantage or disadvantage. The total scores for the projects are summed below in Table 25.

Table 20: Bus tracking screens — advantages and disadvantages

Advantages	**Metric Score**	**Disadvantages**	**Metric Score**
No smartphone needed; serves all ages	Accessibility (+2), Utility (+2)	Hardware at every stop is expensive	Implementation difficulty (-1)
Multilingual UI fits Dublin's commuter mix	Accessibility (+2), Utility (+2)	Screens face vandalism and weather wear	Implementation difficulty (-1), Cost (-2)
Live ETA data cuts wait anxiety	Utility (+2), Market potential (+1)	Needs reliable feeds from operator APIs	Cost (-1), Market potential (-1)
Buyers (bus operator, NTA) are easy to name	Accessibility (+1), Cost (+1), Utility (+2), Market potential (+2)	Power and service contracts add yearly cost	Implementation difficulty (-1), Cost (-2), Market potential (-1)
Heavy daily use in dense corridors	Accessibility (+1), Utility (+1)		

Table 21: Rural transport access — advantages and disadvantages

Advantages	**Metric Score**	**Disadvantages**	**Metric Score**
Clear gap where buses are sparse	Utility (+2), Market potential (+2), Accessibility (+1)	Low density limits fare-box revenue	Cost (-2), Market potential (-1)
Ties to jobs, health visits, and school	Utility (+2), Accessibility (+2)	Weak mobile data complicates digital tools	Accessibility (-2), Implementation difficulty (-1)
Fits EU rural cohesion funds	Market potential (+2), Cost (+1)	Success depends on slow public partnerships	Implementation difficulty (-1), Scalability (-1)
Few direct competitors in the niche	Market potential (+2)	Hard to copy one village model elsewhere	Scalability (-2)

Table 22: Water-based energy — advantages and disadvantages

Advantages	**Metric Score**	**Disadvantages**	**Metric Score**
Cuts reliance on imported fuel	Environmental safety (+1), Implementation difficulty (+1), Accessibility (+2), Cost (+1), Market potential (+1)	Capital cost and timelines exceed the scope of this project	Cost (-2), Market potential (-1), Scalability (-1)
Hydro and tidal tech are mature	Implementation difficulty (+1), Accessibility (+2)	Builds can disturb rivers, coasts, wildlife	Environmental safety (-2), Implementation difficulty (-1)
Long–run payoff and energy security	Utility (+2), Cost (+1), Market Potential (+1)	Permits need national agencies	Implementation Difficulty (-2), Scalability (-1)
Aligns with EU renewable targets	Environmental Safety (+1), Market Potential (+1)	Benefits arrive slowly and spread widely	Utility (-1)
Clear operation after commissioning	Environmental Safety (+1)	Better as policy than as a student prototype	Implementation Difficulty (-2)

Table 23: Corner collision sensors — advantages and disadvantages

Advantages	**Metric Score**	**Disadvantages**	**Metric Score**
Targets a known rural crash pattern	Utility (+2)	Cost scales with every corner equipped	Cost (-1), Scalability (-1)
Simple story for councils and public	Implementation Difficulty (+1), Market Potential (+1)	Remote power is non-trivial	Accessibility (-1)
Strong road-safety procurement angle	Utility (+2)	Outdoor gear needs rugged enclosures	Implementation Difficulty (-1), Scalability (-1)
Proximity sensing and LEDs are off-the-shelf	Cost (+1)	Drivers must learn a new warning meaning	Environmental Safety (-1), Accessibility (-1), Utility (-1)
One corner can be piloted cheaply	Cost (+1), Scalability (+1)	Similar to an earlier mountain road sensor idea	Market Potential (-1)

Table 24: Drain blockage sensor to prevent flooding and standing water — advantages and disadvantages

Advantages	Virtue Explanation	Disadvantages	Virtue Explanation
Early detection/warning	Accessibility(+1), Utility(+2)	Doesn't affect drain capacity	Utility (-1)
Reduced flood damage	Utility(+2), Market Potential(+1)	Initial installation cost	Cost (-2)
Prevents increased pollution	Environmental Safety (+2)	Maintenance & damage control	Implementation Difficulty (-1), Cost (-1)
Improved public safety	Accessibility (+1)	Connection problems	Implementation Difficulty (-1), Accessibility (-1)
Can be done gradually	Market Potential (+1), Scalability (+2)	Power constraints	Implementation Difficulty (-1), Cost (-1)

8.2 Metric Scores

Table 25 lists the total scores of all five projects. Figure 3 plots the totals side by side to visually compare the scores of the projects.

Bus tracking screens rank first at +9 points. The advantages on the bus tracking system outweighed its disadvantages, resulting in a score of +9. This high valuation stems from the project's exceptionally strong scores in accessibility and utility, aligning perfectly with the virtues of community awareness and service. It directly targets commuter anxiety and easily offsets the minor penalties associated with the disadvantages.

Following the bus tracking screens, the rural transport access project ranked second with a total score of +4. This concept scored high in market potential and utility because it fills a major public transport gap for rural residents. However, its final score was lowered by penalties in scalability and implementation difficulty due to low population density and weak regional mobile data.

The water based energy generation and drain blockage sensor concepts tied for third place, each earning a total score of +3. For the drain sensors, strong marks in environmental safety and scalability were dragged down by initial installation costs and implementation difficulties. Similarly, the water energy project consisted of strong accessibility and utility points with implementation and scalability troubles, because hydro and tidal systems are too expensive and complex.

While the project scored well in utility by targeting a known crash pattern, costs quickly add up as more country turns are equipped, and drivers are forced to learn an unfamiliar warning signal. Ultimately, the bus tracking screens remain the best and most practical project to pursue for the rest of the course.

Table 25: Metric scores by project (net sum of advantage and disadvantage points per metric)

Project	Env.	Impl.	Access.	Cost	Util.	Market	Scal.	Total
Bus screens	0	−3	+8	−4	+9	+1	0	**9**
Rural transport	0	−2	+3	−1	+4	+5	−3	**4**
Water energy	0	−3	+4	0	+1	+2	−2	**3**
Corner sensors	−1	−1	−2	+1	+3	0	−1	**0**
Drain sensors	+2	−3	+1	−4	+3	+1	+2	**3**

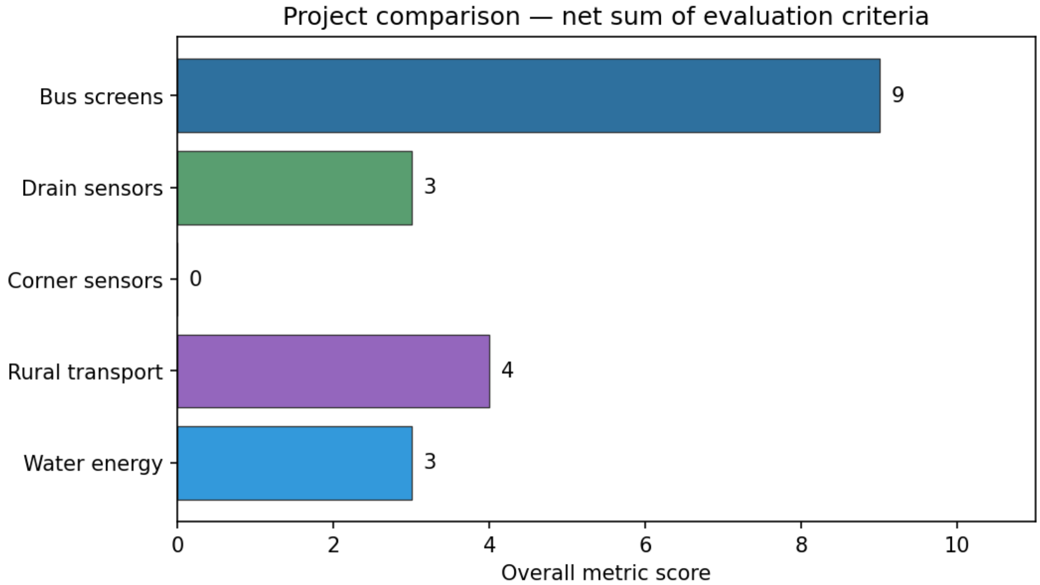


Figure 3: Total evaluation scores for five proposed projects

The spread from 9 to 0 points separates concepts that are ready for a focused course prototype from those better framed as policy or long-horizon infrastructure. The table highlights relative strengths rather than outright failure. Section 6 interprets these results in light of customers and virtue-based trade-offs from Tables 20–24. All of these projects reflect substantial effort and consideration, and the scoring does not imply a preference for one concept beyond its alignment with the selected evaluation criteria. However, given the constraints the team has formed for itself, the project scores show how aligned with the team goals each project is. The highest-scoring project will advance to the next phase of design, unless met with a serious barrier.

8.3 Decision Matrix and Mock Matrix

A decision matrix template was made to be a structured, quantitative tool designed to simplify complex decision-making by eliminating bias and subjectivity. By organizing multiple design alternatives against a predefined set of weighted evaluation criteria, the template allows teams to systematically score and contrast each option. This objective framework calculates total weighted scores to clearly identify the most optimal solution based on performance, constraints, and strategic priorities. This template was created in Microsoft Excel using cell by cell formulas and can be seen in Table 26.

Table 26: Decision matrix template

Issue / Objective:		**Concept Alternatives**							
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma = 1.00$)	**Concept A**		**Concept B**		**Concept C**		**Concept D**	
		Abs. Score	Wgt. Score	Abs. Score	Wgt. Score	Abs. Score	Wgt. Score	Abs. Score	Wgt. Score
Criterion 1:	0	0	0	0	0	0	0	0	0
Criterion 2:	0	0	0	0	0	0	0	0	0
Criterion 3:	0	0	0	0	0	0	0	0	0
Criterion 4:	0	0	0	0	0	0	0	0	0
Weight Validation (must equal 1.00):	0								
Total Weighted Score (Customer Satisfaction Index)	—	—	0	—	0	—	0	—	0

In order to prove effectiveness of the decision matrix template shown in Table 26, a mock decision matrix was created with the issue of deciding where to go to lunch. The comparison criteria were cost, service speed, menu variety, and proximity with the different concepts of pizza, sushi, burgers, and salad. This matrix is shown in Table 27 with the highest customer satisfaction score being 7.2, which is burgers.

Table 27: Mock decision matrix

Issue: Choosing the best place for lunch		**Concept Alternatives**							
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma = 1.00$)	**Pizza**		**Sushi**		**Burgers**		**Salad**	
		Abs. Score	Wgt. Score	Abs. Score	Wgt. Score	Abs. Score	Wgt. Score	Abs. Score	Wgt. Score
Cost (dollars/person):	0.4	8	3.2	4	1.6	7	2.8	5	2
Service speed (minutes):	0.3	7	2.1	3	0.9	8	2.4	9	2.7
Menu variety (number of options):	0.2	5	1	8	1.6	4	0.8	8	1.6
Proximity (miles):	0.1	9	0.9	5	0.5	6	0.6	7	0.7
Weight Validation (must equal 1.00):	1.00								
Total Weighted Score (Customer Satisfaction Index)	—	—	7.2	—	4.6	—	6.6	—	7

8.4 Concept Selection and Decision Matrix

To select the preferred multilingual RTPI display concept, the team developed a decision matrix shown in Tables 28 through 35 using the functional concepts generated in Section 6 and the quantitative models established in Section 6.2. The concepts evaluated included button based language selection, multilingual audio announcements, pictogram displays, automatic language cycling, and simultaneous multilingual display, each represented by multiple design variables as stated in Section 6.2.

The importance/weighting of each comparison criteria was determined by referencing the ranked engineering specifications established in Section 5. Since Compliance with NTA Accessibility

Standards was identified as one of the highest priority engineering specifications, NTA Accessibility Compliance received the largest weighting of 0.45. Screen Legibility was assigned a weight of 0.30 to reflect the importance of providing information that can be quickly and accurately interpreted. Multilingual Support received a weight of 0.25, recognizing the project's goal of serving tourists and non-English speakers while balancing this objective against readability and accessibility. These normalized weights sum to 1.00 and preserve the relative priorities established through the earlier customer requirements and engineering specification rankings.

For each design alternative, the associated mathematical models were used to calculate the shared state variables, which were then converted into normalized measures of merit scores. The weighted scores for each criteria were summed to produce an overall customer satisfaction index for every concept, enabling an objective comparison of the alternatives.

The completed decision matrix identified Pictogram 4 (detailed in Table 32) as the highest-performing engineering concept, achieving the maximum overall customer satisfaction index of 8.34. This concept scores exceptionally well due to its superior accessibility compliance and strong screen legibility, utilizing universally understood visual markers to provide efficient multilingual support without requiring passengers to physically interact with the display. Consequently, the team has selected the Pictogram-based display framework as the primary functional concept to pursue. This concept provides the optimal balance between universal accessibility, rapid readability, and inclusive communication according to the prioritized engineering specifications and measures of merit established throughout the design process.

Table 28: Comparison Matrix – Buttons 1 to 4

		Button 1 (300, 50, F14)		**Button 2** (300, 50, F16)		**Button 3** (300, 60, F14)		**Button 4** (300, 60, F16)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	3.00	1.35	7.00	3.15	6.00	2.71	10.00	4.53
2. Screen Legibility	0.30	6.83	2.05	8.17	2.45	6.83	2.05	8.17	2.45
3. Multilingual Support	0.25	1.82	0.45	1.82	0.45	0.91	0.23	0.91	0.23
Total Weighted Score (CSI)	**1.00**	–	**3.85**	–	**6.05**	–	**4.98**	–	**7.18**

Table 29: Comparison Matrix – Buttons 5 to 8

		Button 5 (400, 50, F14)		**Button 6** (400, 50, F16)		**Button 7** (400, 60, F14)		**Button 8** (400, 60, F16)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	3.00	1.35	7.00	3.15	6.00	2.71	10.00	4.53
2. Screen Legibility	0.30	6.83	2.05	8.17	2.45	6.83	2.05	8.17	2.45
3. Multilingual Support	0.25	0.91	3.64	0.91	3.64	2.42	0.61	2.42	0.61
Total Weighted Score (CSI)	**1.00**	–	**4.31**	–	**6.51**	–	**5.36**	–	**7.56**

Table 30: Comparison Matrix – Audio 1 to 4

		Audio 1 (M100, L10, Q80)		Audio 2 (M100, L10, Q90)		Audio 3 (M100, L15, Q80)		Audio 4 (M100, L15, Q90)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	4.00	1.80	7.00	3.15	4.00	1.80	7.00	3.15
2. Screen Legibility	0.30	5.83	1.75	5.83	1.75	5.83	1.75	5.83	1.75
3. Multilingual Support	0.25	5.45	1.36	5.45	1.36	2.42	0.61	2.42	0.61
Total Weighted Score (CSI)	**1.00**	–	**4.91**	–	**6.26**	–	**4.16**	–	**5.51**

Table 31: Comparison Matrix – Audio 5 to 8

		Audio 5 (M150, L10, Q80)		Audio 6 (M150, L10, Q90)		Audio 7 (M150, L15, Q80)		Audio 8 (M150, L15, Q90)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	4.00	1.80	7.00	3.15	4.00	1.80	7.00	3.15
2. Screen Legibility	0.30	5.83	1.75	5.83	1.75	5.83	1.75	5.83	1.75
3. Multilingual Support	0.25	10.00	2.50	10.00	2.50	5.45	1.36	5.45	1.36
Total Weighted Score (CSI)	**1.00**	–	**6.05**	–	**7.40**	–	**4.91**	–	**6.26**

Table 32: Comparison Matrix – Pictograms 1 to 4

		Pictogram 1 (Icons:10, S10)		Pictogram 2 (Icons:10, S15)		Pictogram 3 (Icons:20, S10)		Pictogram 4 (Icons:20, S15)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	2.50	1.13	2.50	1.13	10.00	4.50	10.00	4.50
2. Screen Legibility	0.30	4.17	1.25	7.50	2.25	4.17	1.25	7.50	2.25
3. Multilingual Support	0.25	1.82	0.45	1.82	0.45	6.36	1.59	6.36	1.59
Total Weighted Score (CSI)	**1.00**	–	**2.83**	–	**3.83**	–	**7.34**	–	**8.34**

Table 33: Comparison Matrix – Cycling 1 to 4

		Cycling 1 (Wait:60, Cyc:10)		Cycling 2 (Wait:60, Cyc:15)		Cycling 3 (Wait:120, Cyc:10)		Cycling 4 (Wait:120, Cyc:15)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	5.00	2.25	5.00	2.25	5.00	2.25	5.00	2.25
2. Screen Legibility	0.30	4.17	1.25	4.17	1.25	4.17	1.25	4.17	1.25
3. Multilingual Support	0.25	1.82	0.45	0.00	0.00	7.27	1.82	3.64	0.91
Total Weighted Score (CSI)	**1.00**	–	**3.95**	–	**3.50**	–	**5.32**	–	**4.41**

Table 34: Comparison Matrix – Simultaneous 1 to 4

		Simultaneous 1 (Scr:300, F20, R3)		Simultaneous 2 (Scr:300, F20, R4)		Simultaneous 3 (Scr:300, F25, R3)		Simultaneous 4 (Scr:300, F25, R4)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	4.00	1.80	4.00	1.80	9.00	4.05	9.00	4.05
2. Screen Legibility	0.30	7.50	2.25	7.50	2.25	10.00	3.00	10.00	3.00
3. Multilingual Support	0.25	0.91	0.23	0.00	0.00	0.00	0.00	0.00	0.00
Total Weighted Score (CSI)	**1.00**	–	**4.28**	–	**4.05**	–	**7.05**	–	**7.05**

Table 35: Comparison Matrix – Simultaneous 5 to 8

		Simultaneous 5 (Scr:400, F20, R3)		Simultaneous 6 (Scr:400, F20, R4)		Simultaneous 7 (Scr:400, F25, R3)		Simultaneous 8 (Scr:400, F25, R4)	
Comparison Criteria (State Variable & Units)	**Weight** ($\Sigma =$ 1.00)	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score	Abs. Score	Wt. Score
1. NTA Compliance	0.45	4.00	1.80	4.00	1.80	9.00	4.05	9.00	4.05
2. Screen Legibility	0.30	7.50	2.25	7.50	2.25	10.00	3.00	10.00	3.00
3. Multilingual Support	0.25	2.42	0.61	0.91	0.23	1.21	0.30	0.00	0.00
Total Weighted Score (CSI)	**1.00**	–	**4.66**	–	**4.28**	–	**7.35**	–	**7.05**

9 Discussion

9.1 Project Selection & Analysis of Pros and Cons

Bus tracking screens lead with a total score of +9. The concept scores highest on accessibility, utility, and market potential—the criteria tied directly to customers named in Section 1. Rural transport access places second with a score of +4 against the metrics. Limited technological infrastructure in rural areas, lack of public partnerships, and the unique characteristics of small villages complicate implementation, putting it below bus tracking screens. Water-based energy and drainage blockers tied for third with a score of +3. For both concepts, the large-scale implementation challenges and high costs reduced their overall scores, and put these at third. Lastly, corner collision sensors scored a 0 because the advantages and disadvantages were even according to our metrics. Implementation challenges and the need for drivers to learn a new warning system contributed to this concept receiving the lowest score.

No concept is without limitations. Bus screens still need API reliability and vandal-resistant enclosures. Drain sensors monitor blockages but do not increase drain capacity. Corner sensors need a field power plan. The matrix should be viewed as a structured filter rather than a final build order.

9.2 Virtues, Customer Requirements & Engineering Specification Verification

The design chosen in Part 1, bus tracking screens, relies on the research base in Section 2 and a translation of virtues into customer requirements and then into quantifiable engineering specifications. Virtues such as inclusivity, community awareness, and respect directly shaped the prioritized customer requirements, ensuring the system serves vulnerable populations like people with disabilities, non-English speakers, tourists, and commuters without smartphones. Similarly, the virtue of sustainability drove the requirements for weather resistance and low maintenance to maximize the system's operational lifespan and support local sustainability goals.

To convert the customer requirements to engineering specifications, these qualitative customer requirements were mapped onto quantifiable engineering specifications. While customer requirements are ranked by user value, the engineering specifications are organized by system performance and feasibility to ensure the technical architecture can actually deliver on those promises. For example, the top-ranked need for accuracy was translated into a measurable target of ≤ 25 % error between predicted and actual arrival times. Furthermore, high-level accessibility and multilingual goals were given hard targets, such as a compliance score of ≥ 90 against NTA accessibility standards and support for ≥ 4 languages.

The engineering specifications in Table 7 were intentionally written in measurable terms so that they can be verified during testing and implementation. Accuracy can be verified by comparing predicted bus arrival times to actual arrival times during field trials and calculating the percent error. Accessibility requirements can be verified through compliance audits based on NTA accessibility standards and user testing with individuals who have disabilities. Weather resistance can be confirmed through environmental testing and certification of the display enclosure's IP rating. Data refresh rates, delay notifications, and service alerts can be verified by measuring the time between updates in the transit data feed and their appearance on the display. Similarly, multilingual support, route display capacity, and city-wide coverage can be evaluated through functional testing and deployment assessments. These verification methods provide objective evidence that the final design satisfies both the engineering specifications and the customer requirements identified throughout the project.

9.3 Quantitative Evaluation Framework for the Bus Tracking Screen

The qualitative requirements established in preceding phases were operationalized into a quantitative evaluation framework consisting of an augmented engineering specification set grounded in value-sensitive design (VSD), a comprehensive Measures of Merit (MoM) schema, a MATLAB normalization script, and a subsystem decision-matrix workbook. While prior work defined high-level stakeholder mandates, this phase establishes mathematical metrics to evaluate and compare physical design alternatives during detailed engineering design.

9.3.1 Value-Sensitive Design and Ethical Augmentations

To maintain stakeholder trust and institutional accountability, the baseline engineering specifications were re-evaluated through a VSD lens, prioritizing *Reliability* and *Inclusivity* as core functional constraints:

- **System Reliability:** Formulated around the requirement for verifiable, accurate real-time passenger information (RTPI) independent of personal mobile devices. Addressing empirical data on passenger confidence drops caused by "ghost buses," a new metric was synthesized: Ghost Bus Discrepancy Reporting Error Rate (E_{ghost}), bounded at an operational target of $\leq 5\%$. This metric consolidates data-integrity vectors into a single auditable constraint positioned at Rank 2 within the performance hierarchy, immediately below core safety compliance.

- **Interface Inclusivity:** Validated outdoor legibility and multi-user equity metrics rather than adding redundant criteria. Confirmed parameters demand native support for at least four languages, minimum text height thresholds, and high-contrast performance suitable for dynamic ambient lighting environments.

- **Public Stewardship:** Four additional bounding metrics were integrated into the design envelope to prevent sub-optimization from silently omitting ethical considerations: GDPR-aligned encryption privacy (P_{gdpr}), structural vandalism downtime resiliency (t_{down}), material component recyclability (M_{recycle}), and predictive routing algorithmic fairness (I_{trans}).

9.3.2 Workbook Architecture and Downstream Execution

Verification of the decision-matrix workbook architecture Part 3 - Decision Matrix Template.xlsx was achieved using an isomorphic mock model to validate the underlying spreadsheet mechanics, ensuring that criteria weights strictly sum to 1.00 and that total customer satisfaction indexes derive exclusively from weighted product scores.

For the hardware implementation phase, the template segregates evaluation into discrete subsystem matrices—specifically display technology, enclosure structures, and RTPI client architectures. Subsystem evaluation criteria are isolated to the highest-ranking MoMs relevant to that technical domain (e.g., E_{ghost}, CR_{leg}, R_{IP}, and T_{int} for display selection). Column weights remain unassigned until functional models are finalized, ensuring weighting distributions reflect empirical subsystem priorities rather than early analytical assumptions. Moving forward, estimated state variables from physical concepts will be compiled through `calculateMoM` to evaluate alternatives directly against top-tier targets: ghost-bus suppression, honest state labeling, and multilingual accessibility.

9.4 Concept Selection Justification

9.4.1 Synthesis of Selection and Metric Mapping

The transition from broad concept generation to a single, mathematically justified design requires a rigorous mapping of qualitative customer requirements to quantitative engineering specifications. By evaluating the generated alternatives against the established Measures of Merit (MoMs), the selection process bypassed subjective bias in favor of data-driven optimization. While initial brainstorming in this engineering design step yielded diverse approaches—ranging from automatic language cycling to interactive user interfaces—the parametric sweeps executed via the MATLAB scoring framework revealed distinct performance thresholds across the multilingual display subsystem.

[Customer Requirements] $\rightarrow$ [Engineering Specs / MoMs] $\rightarrow$ [Parametric MATLAB Sweep] $\rightarrow$ [Optimized Selection]

The selection of Pictogram 4 (achieving the peak customer satisfaction index of 8.34) directly satisfies the core customer requirement of universal, rapid-glance readability without relying on physical passenger interaction. This is critical for high-throughput public transit environments where manual display manipulation introduces physical accessibility barriers and delays.

9.4.2 Alignment with Engineering Specifications, MoMs, and Virtues

The superiority of the pictogram-based framework is rooted in its mathematically proven balance across competing design metrics. With respect to engineering specifications and MoMs, language cycling concepts inherently suffered from lower legibility scores due to the temporal delay of rotating text, whereas Pictogram 4 maximizes information density simultaneously. It satisfies the strict screen legibility thresholds by utilizing standardized visual markers that bypass language barriers entirely, keeping message processing times well within the target limits defined in the engineering specifications.

With respect to value sensitive design and virtues, this selection extends beyond raw performance numbers. It embodies the core engineering virtues of Safety and Inclusivity outlined in the Value Sensitive Design (VSD) framework and the NSPE Code of Ethics. By prioritizing an interface that does not require literacy in a specific language or physical contact with the hardware, the design honors the ethical imperative to provide equitable public utility access to all demographics, including tourists, foreign-language speakers, and mobility-impaired passengers.

9.4.3 Methodological Necessity of Considering All Concepts Before Evaluation

Executing extensive design-variable sweeps and organizing the results in a multi-matrix format was a necessary step in this engineering design stage to rigorously stress-test the sensitivity of each concept. Without sweeping parameters across varying operational bounds, a concept might appear optimal under ideal conditions but fail under strict pass/fail constraints. The workflow implemented in `evaluationConceptMain.m`—collecting state-variable arrays across all tested variants, setting SV_{min} from engineering specifications, setting $SV_{max} = \max(SV_{array})$ from the sweep, and normalizing through `calculateMoM.m`—ensures that each alternative is scored on a common 0–10 scale before entering the decision matrix. This iterative modeling ensures that the chosen configuration behaves predictably and robustly across the full tradespace rather than at a single hand-picked design point.

References

[1] National Transport Authority, "Public Service Obligation Performance Annual Report 2024," Tech. rep., National Transport Authority, Dublin, Ireland, 2025, accessed May 30, 2026.

[2] National Transport Authority, "NTA Awards Bus AVL Contract to Trapeze ITS UK Ltd." online, 2023, accessed June 7, 2026.

[3] European Parliament and Council of the European Union, "Regulation (EU) No 181/2011 concerning the rights of passengers in bus and coach transport," online, 2011, accessed June 7, 2026.

[4] Government of Ireland, “European Union (Accessibility Requirements for Products and Services) Regulations 2023 (S.I. No. 636 of 2023),” online, 2023, accessed June 7, 2026.
[5] National Transport Authority, “Direct Award Bus Service Contracts: Schedules (DBDA),” online, 2025, Schedule 14 RTPI obligations; accessed June 7, 2026.
[6] National Transport Authority, “Responses to Parliamentary Questions Bulletin 5, Q2 2025,” Tech. rep., National Transport Authority, Dublin, Ireland, 2025, Ghost bus and RTPI accuracy; accessed June 7, 2026.
[7] Trustpilot, “Dublin Bus Customer Reviews,” online, 2026, accessed June 7, 2026.
[8] Transport for Ireland, “TFI Live App,” online, 2026, accessed June 7, 2026.

Appendix A - MATLAB Code

Listing 1: MATLAB script (scoring$_{c}$*omparison.m)thatgeneratestheprojectscorebarchart.*

```
projects = {'Bus screens', 'Drain sensors', 'Corner sensors', ...
            'Rural transport', 'Water energy'};
scores = [9, 3, 0, 4, 3];

figure('Color', 'w', 'Position', [100 100 720 420]);
b = barh(scores);
b.FaceColor = 'flat';
b.CData = [0.18 0.44 0.62;   % bus  blue
           0.35 0.62 0.44;   % drain  green
           0.82 0.45 0.20;   % corner  orange
           0.58 0.40 0.74;   % rural  purple
           0.20 0.60 0.86];  % water  light blue
b.EdgeColor = [0.2 0.2 0.2];
b.LineWidth = 0.6;

set(gca, 'YTick', 1:5, 'YTickLabel', projects, 'YDir', 'reverse');
xlabel('Overall metric score');
title('Project comparison  net sum of evaluation criteria');
xlim([0 max(scores) + 2]);

for k = 1:numel(scores)
    text(scores(k) + 0.15, k, num2str(scores(k)), ...
        'VerticalAlignment', 'middle', 'FontSize', 10);
end

outFile = fullfile(getenv('HOME'), 'Desktop', 'scoring_comparison.png');
saveas(gcf, outFile);
disp(['Saved ', outFile]);
```

Listing 2: MATLAB script (`calculateMoM.m`) that normalizes a state variable to a 0–10 measure-of-merit score from minimum and maximum bounds.

```
function mom_score = calculateMoM(state_val, min_val, max_val, direction)
% Calculates Measures of Merit
% Inputs:
%   state_val - The specific state variable value
%   min_val   - The minimum value constraint
%   max_val                 - The maximum value constraint
%   direction - A string indicating preference: 'higher' if larger values...
%   are better and 'lower'  if smaller values are better
% Outputs:
%   mom_score - A scale (0 worst, 10 best)

    % Calculate the Measure of Merit based on direction
    if max_val == min_val
        error('Maximum and minimum values cannot be identical.');
        return
    end

    % Check direction preference and apply engineering specifications up front
    switch lower(direction)
        case 'higher'
            if state_val <= min_val
                mom_score = 0;
```

```
            elseif state_val >= max_val
                mom_score = 10;
            else
                mom_score = 10 * ((state_val - min_val) / (max_val - min_val));
            end

        case 'lower'
            if state_val >= max_val
                mom_score = 0;
            elseif state_val <= min_val
                mom_score = 10;
            else
                mom_score = 10 * ((max_val - state_val) / (max_val - min_val));
            end

        otherwise
            error('Error. Use higher or lower.');
            return
    end
end
```

Listing 3: MATLAB script (`evaluationConceptMain.m`) that sweeps design variables for all five multilingual display concepts, collects state variables into master arrays, sets dynamic maximum bounds from the full tradespace sweep.

```
clc;
clear;

%% Fixed Engineering Minimum Thresholds (From Eng. Specs)
xmin = [4 3 90]; % [SV1_min, SV2_min, SV3_min]

%% =====================================================
%% PHASE 1: GENERATE AND COLLECT ALL RAW VALUES
%% =====================================================

% --- 1. Touchscreen Button Concept ---
A_s_btn = [300 400]; A_b_btn = [50 60]; F_s_btn = [14 16];
raw_SV1_btn = []; raw_SV2_btn = []; raw_SV3_btn = [];
for i = 1:length(A_s_btn)
    for j = 1:length(A_b_btn)
        for k = 1:length(F_s_btn)
            [N, CR, A] = buttonConcept2(A_s_btn(i), A_b_btn(j), F_s_btn(k));
            raw_SV1_btn = [raw_SV1_btn, N];
            raw_SV2_btn = [raw_SV2_btn, CR];
            raw_SV3_btn = [raw_SV3_btn, A];
        end
    end
end

% --- 2. Audio Announcements Concept ---
M_a_aud = [100 150]; S_l_aud = [10 15]; Q_a_aud = [80 90];
raw_SV1_aud = []; raw_SV2_aud = []; raw_SV3_aud = [];
for i = 1:length(M_a_aud)
    for j = 1:length(S_l_aud)
        for k = 1:length(Q_a_aud)
            [N, CR, A] = audioConcept2(M_a_aud(i), S_l_aud(j), Q_a_aud(k));
            raw_SV1_aud = [raw_SV1_aud, N];
            raw_SV2_aud = [raw_SV2_aud, CR];
```

```
            raw_SV3_aud = [raw_SV3_aud, A];
        end
    end
end

% --- 3. Pictograms Concept ---
N_p_pic = [10 20]; I_s_pic = [10 15];
raw_SV1_pic = []; raw_SV2_pic = []; raw_SV3_pic = [];
for i = 1:length(N_p_pic)
    for j = 1:length(I_s_pic)
        [N, CR, A] = pictogramConcept2(N_p_pic(i), I_s_pic(j));
        raw_SV1_pic = [raw_SV1_pic, N];
        raw_SV2_pic = [raw_SV2_pic, CR];
        raw_SV3_pic = [raw_SV3_pic, A];
    end
end

% --- 4. Automatic Cycling Concept ---
T_w_cyc = [60 120]; T_c_cyc = [10 15];
raw_SV1_cyc = []; raw_SV2_cyc = []; raw_SV3_cyc = [];
for i = 1:length(T_w_cyc)
    for j = 1:length(T_c_cyc)
        [N, CR, A] = cyclingConcept2(T_w_cyc(i), T_c_cyc(j));
        raw_SV1_cyc = [raw_SV1_cyc, N];
        raw_SV2_cyc = [raw_SV2_cyc, CR];
        raw_SV3_cyc = [raw_SV3_cyc, A];
    end
end

% --- 5. Simultaneous Display Concept ---
A_s_sim = [300 400]; F_s_sim = [20 25]; N_r_sim = [3 4];
raw_SV1_sim = []; raw_SV2_sim = []; raw_SV3_sim = [];
for i = 1:length(A_s_sim)
    for j = 1:length(F_s_sim)
        for k = 1:length(N_r_sim)
            [N, CR, A] = simultaneousConcept2(A_s_sim(i), F_s_sim(j), N_r_sim(k));
            raw_SV1_sim = [raw_SV1_sim, N];
            raw_SV2_sim = [raw_SV2_sim, CR];
            raw_SV3_sim = [raw_SV3_sim, A];
        end
    end
end

%% =====================================================
%% PHASE 2: COMBINE ALL RAW VALUES INTO SINGLE MASTER ARRAYS
%% =====================================================

SV1_raw_master = [raw_SV1_btn, raw_SV1_aud, raw_SV1_pic, raw_SV1_cyc, raw_SV1_sim];
SV2_raw_master = [raw_SV2_btn, raw_SV2_aud, raw_SV2_pic, raw_SV2_cyc, raw_SV2_sim];
SV3_raw_master = [raw_SV3_btn, raw_SV3_aud, raw_SV3_pic, raw_SV3_cyc, raw_SV3_sim];

% Find global maximums across all options as explained in professor sketch
xmax = [max(SV1_raw_master), max(SV2_raw_master), max(SV3_raw_master)];

disp('------------------------------------------------------------------')
disp('PHASE 1: MASTER ARRAYS BEFORE MEASURE OF MERIT CALCULATIONS (RAW VALUES)')
disp('------------------------------------------------------------------')
```

```
fprintf('\nGlobal Minimum Bounds (From Specs): SV1_min = %.2f, SV2_min = %.2f, SV3_min
     = %.2f\n', xmin(1), xmin(2), xmin(3));
fprintf('Calculated Global Maximum Bounds:    SV1_max = %.2f, SV2_max = %.2f, SV3_max
    = %.2f\n\n', xmax(1), xmax(2), xmax(3));

disp('SV1 Master Array (Raw Languages Support Counts):');
disp(SV1_raw_master);
disp('SV2 Master Array (Raw Legibility Contrast Ratios):');
disp(SV2_raw_master);
disp('SV3 Master Array (Raw Accessibility Compliance Scores):');
disp(SV3_raw_master);


%% ========================================================
%% PHASE 3: COMPUTE GLOBAL MoM SCORES (ROW-ORIENTED FOR COPY-PASTE)
%% ========================================================

SV1_mom_master = zeros(size(SV1_raw_master));
SV2_mom_master = zeros(size(SV2_raw_master));
SV3_mom_master = zeros(size(SV3_raw_master));

% Loop through the single master arrays to call the normalization function
for i = 1:length(SV1_raw_master)
    SV1_mom_master(i) = measureOfMerit2(SV1_raw_master(i), xmin(1), xmax(1), 1);
end

for i = 1:length(SV2_raw_master)
    SV2_mom_master(i) = measureOfMerit2(SV2_raw_master(i), xmin(2), xmax(2), 1);
end

for i = 1:length(SV3_raw_master)
    SV3_mom_master(i) = measureOfMerit2(SV3_raw_master(i), xmin(3), xmax(3), 1);
end

disp('-----------------------------------------------------------------------')
disp('PHASE 2: FINAL MEASURE OF MERIT ARRAYS (0-10 SCORES FOR DECISION MATRIX ROWS)')
disp('-----------------------------------------------------------------------')
disp('Instructions: Each array below represents a continuous row in your final matrix
    .')
disp('The elements correspond sequentially to the configuration columns of your
    options.')
disp(' ')

disp('SV1 Final MoM Row (Languages Supported Scores):');
disp(SV1_mom_master);

disp('SV2 Final MoM Row (Legibility Scores):');
disp(SV2_mom_master);

disp('SV3 Final MoM Row (Accessibility Compliance Scores):');
disp(SV3_mom_master);
```

Listing 4: MATLAB function (`buttonConcept2.m`) that maps button design variables to shared state variables.

```
function [N_lang, CR_leg, A_nta] = buttonConcept2(A_s,A_b,F_s)

N_lang = A_s/A_b;
```

```
CR_leg = 0.8*F_s;
A_nta = 50 + 0.3*A_b + 2*F_s;

end
```

Listing 5: MATLAB function (`audioConcept2.m`) that maps audio announcement design variables to shared state variables.

```
function [N_lang, CR_leg, A_nta] = audioConcept2(M_a,S_l,Q_a)

N_lang = M_a/S_l;
CR_leg = 10;
A_nta = 70 + 0.3*Q_a;

end
```

Listing 6: MATLAB function (`pictogramConcept2.m`) that maps pictogram design variables to shared state variables.

```
function [N_lang, CR_leg, A_nta] = pictogramConcept2(N_p,I_s)

N_lang = 1 + 0.5*N_p;
CR_leg = 0.8*I_s;
A_nta = 85 + 0.75*N_p;

end
```

Listing 7: MATLAB function (`cyclingConcept2.m`) that maps automatic cycling design variables to shared state variables.

```
function [N_lang, CR_leg, A_nta] = cyclingConcept2(T_w,T_c)

N_lang = T_w/T_c;
CR_leg = 8;
A_nta = 95;

end
```

Listing 8: MATLAB function (`simultaneousConcept2.m`) that maps simultaneous display design variables to shared state variables.

```
function [N_lang, CR_leg, A_nta] = simultaneousConcept2(A_s,F_s,N_r)

N_lang = A_s/(F_s*N_r);
CR_leg = 0.6*F_s;
A_nta = 74 + F_s;

end
```

Appendix B - Team Contract

Team Contract

Team Name/Number: 1 | **Date: June 3, 2026**

Team Member	Signature
Ava Towery	
Sarah Senn	
Tyler Asmussen	
Odon Ineza	

Team Performance Expectations

- Through our methods of communication, all team members will share their times of availability. A minimum of two meeting times will be decided for each project part before its respective submission date. If there is no overlapping free time, the group will choose the time that works for the majority of members and ensure that absent members are updated thoroughly afterward.
- All team members agree to be on time to meetings and respect each other's time and schedules.
- Each team member should have their portion of the assignment completed and reviewed by themselves at least 8 hours prior to the official due date.
- At least 8 hours before the due date, the team should begin an overall review of the project documents and deliverables to ensure completeness, accuracy, organization, and professionalism. Team members will review one another’s work and provide constructive feedback if needed.
- Each member is expected to actively participate in discussions, brainstorming, planning, and decision-making processes throughout the duration of the project.
- During meetings, team members are expected to remain attentive, respectful, and engaged in discussion. Side conversations, excessive phone use, and other distractions should be minimized to maintain productivity and respect for everyone’s time.
- Each member should research and evaluate their ideas before presenting them to the group in order to minimize time spent discussing unrealistic or non-feasible ideas.
- Workload will be evenly distributed, and to ease difficulty, if an individual feels behind or needs guidance, all members are open to helping and collaborating on multiple parts of the project.
- All members are expected to communicate honestly regarding progress, scheduling conflicts, or difficulties completing assigned tasks as early as possible.

Member Roles:

- At each meeting, a different team member will take meeting minutes, with responsibilities rotating in the following order: Odon, Sarah, Ava, and Tyler.

- Meeting minutes, task assignments, deadlines, important decisions, and project revisions will be documented and stored in shared folders through Google Drive and Overleaf to ensure organization, accountability, and accessibility for all team members.
- Roles and responsibilities will be distributed during each meeting based primarily on confidence, experience, availability, and individual interest in exploring new areas.
- If there is a task that is unfamiliar to all team members and no one feels confident completing it individually, a meeting will be scheduled to collaboratively work through the task as a team.
- All members are expected to contribute fairly to brainstorming, research, documentation, presentations, and project development regardless of assigned roles.

Strategies for Conflict Resolution:

- Team decisions will ideally be made through consensus. If consensus cannot be reached, decisions will be made by majority vote, with each member having one equal vote. If a majority vote cannot be reached, the team will reach out to the PAs and/or professor for advice and guidance.
- Team members agree to address conflicts by listening respectfully, avoiding assumptions about intent, and focusing on solutions and future improvement rather than assigning blame.
- If a team member repeatedly fails to meet expectations, misses deadlines, fails to communicate, or violates the team contract, the issue will first result in a warning and discussion within the group. Continued issues may be documented and brought to the PAs, and if the situation cannot be solved, the issue will be brought to the professor for mediation if necessary.
- All conflicts and disagreements should be handled respectfully and professionally without personal attacks, hostility, or disrespectful communication.

Communication Norms:

- The primary methods of communication will be the team group chat, email, and shared collaborative platforms via Google Drive and Overleaf (LaTeX).
- Meetings will be scheduled through the team group chat and agreed upon based on all members' availability. Any schedule changes should be communicated at least 24 hours in advance whenever possible.
- Communication should remain respectful, professional, and constructive at all times, even during disagreements or stressful situations.
- Members should notify the group as soon as possible if they anticipate missing a meeting, being late, or having difficulty completing assigned work.
- Team members are expected to respond to important messages within a reasonable amount of time to maintain effective communication and workflow.
- Non-urgent communication should generally occur between 8:00 AM and 10:00 PM out of respect for team members' personal schedules, responsibilities, well-being, and time for evening outings and activities.
- By signing this contract, all team members agree to uphold the expectations, responsibilities, and professional standards outlined above throughout the duration of the project.

Appendix C - Meeting Minutes

Team Meeting Minutes		
Design Organization: Group 1		**Date: Thursday May 28, 2026**
Agenda: 1. The team has met to set the ground rules for the team and formulate a team contract that they will uphold during the course of the project. 2. Discussing the next steps and workload distribution for Part 1 of the project. 3. Coming up with a list of possible customers and associated projects.		
Discussion: 1. In today's discussion, the team talked about the performance and expectations they will be held to during this project. Values like communication, punctuality, open mindedness, working ahead of schedule, thoroughness, and even distribution of tasks were agreed upon as common ground for all the members. Regarding performance and expectations, the team has agreed to dedicate the fullest of their attention to the perfecting of this project, and in the case that change is needed, the team has allowed to leave the contract as a live document— meaning that in the case change occurs and the revision of the contract is needed, the team will come back to the contract and discuss the necessary changes. 2. The team also discussed how the tasks will be allocated within the group and some metrics were set regarding how the tasks will be evenly distributed according to individual strengths within the team. 3. The team went over strategies for conflict resolution where they agreed upon the decision making process and how they will go about voting, handling a team member who consistently doesn't meet the expectations. 4. Last thing the team discussed on the contract, they talked about the communication modes and norms. First the team established convenient modes of communication, then they moved onto the step of setting norms about how these channels will be used and when they will be used. 5. Brainstormed 5 ideas for alternative projects.		
Decisions Made: 1. All meeting times for the first deliverable were established. 2. All tasks for the first deliverable were distributed. 3. All alternative projects were considered and added to the alternative projects document. 4. Narrowed down 5 ideas for the project.		
Action Items	Person Responsible	Deadline
Brain storm ideas for customers and projects.	All	Monday

Make a pros and cons list for all ideas.	All	Tuesday

Team member:	Ava Towery	Date for next meeting:
Team member:	Sarah Senn	Tuesday June 2, 2026
Team member:	Tyler Asmussen	Wednesday June 3, 2026
Team member:	Odon Ineza	

Team Meeting Minutes

Design Organization: Group 1	**Date: Tuesday June 2, 2026**

Agenda:
1. Review part 1 of the project – pro/con lists, ideas of more potential customers and ideas, and formatting.
2. Assign work that is left for part 1

Discussion:
For project 1, we have to add virtues for the pro/con lists and clarify the different scales for the tables. Potential reference for the Dublin bus system since that is the idea that seems to be the strongest at this point.

Some ideas about bus tracking screens: The idea of having an interactive path with all buses on a screen, but that may be too busy and requires a lot of electricity. Idea about having a picture of the route map with a dot that lights up where the bus currently is to conserve electricity. Language barriers were also discussed and the idea of having phrases with different languages cycle through to make the "screen" more accessible to all.

Add an idea to part 1 – ideas about Dublin traffic, older homes, and flooding were presented. The idea is to fix flooding with a sensor that tells people that the drains are clogged and need to be cleaned out to hopefully allow the drains to work and prevent flooding.

Decisions Made:
1. Find a data set that goes with the need for bus tracking screens
2. Virtues must be added to pro/con tables
3. Add flooding idea to the list of potential customers and projects and create a pro/con list for that

Action Items	Person Responsible	Deadline

Find reference for bus system issue and cite	Tyler	June 3
Add virtues to pro/con lists	All	June 3
Add flooding idea to part 1	Ava	June 3
Create appendix	All	June 3

Team member:	Ava Towery	Date for next meeting:
Team member:	Sarah Senn	Wednesday, June 3, 2026
Team member:	Tyler Asmussen	
Team member:	Odon Ineza	

Team Meeting Minutes

Design Organization:	**Group 1**	**Date: Thursday, June 4, 2026**

Agenda:
1. Discuss upcoming roles based of Final Project Part 2
2. Begin finding websites for both customer requirements and engineering specifications

Discussion:
In today's class meeting, we discussed the next steps for our project. Having already decided to proceed with the bus tracking screens, we focused on identifying standard requirements and customer values. Our goal is to develop a prioritized list of project requirements, ensuring that the most important user needs and functional features are addressed first. This list will guide the design and development process moving forward.

Moving forward we will split up the two main components of part 2, defining customer requirements and engineering specifications. We decided it was best for everyone to look for reviews and requirements and then once that is complete the two assigned to customer requirements can complete the full list. After that the remaining two team members can complete their portion which is translating these requirements to engineering specifications. After this the team will meet again to overall review the document.

In order to see what needs to be done we found information on the tracking systems already installed on the busses. Currently there are tracking systems on the bus but they aren't accessible to the general public. This is helpful because it means that trackers would have to be installed but rather a policy would have to be passed to access the trackers on the bus and connect them to the screens. Currently when tracking the bus passengers must be on the transit app and then other customers can see where the bus is actively moving based on the passengers location. This is inaccurate and relies on others to be on their phone and to track them rather than the bus itself.

Decisions Made:		
1. Everyone should find reviews and customer preferences 2. Everyone should find legal requirements necessary for the project 3. Sarah and Ava will then complete the customer requirements portion 4. Then Odon and Tyler can complete the engineering specifications 5. After notes are taken in google docs the team will transfer it in a proper format to overleaf		
Action Items	Person Responsible	Deadline
Find customer reviews to similar products.	Everyone	June 5
Find standard legal requirements.	Everyone	June 5
Define customer requirements	Sarah and Ava	June 6
Translate customer requirements to engineering specifications	Odon and Tyler	June 7

Team member:	Ava Towery	Date for next meeting:
Team member:	Sarah Senn	Sunday June 7 2026
Team member:	Tyler Asmussen	
Team member:	Odon Ineza	

The Mechanical Design Process

Team Meeting Minutes

Design Organization: Group 1	**Date: Tuesday June 9, 2026**

Agenda:
1. Review feedback from final project part 2
2. Finalize customer requirements and engineering specifications
3. Discuss the implementation of virtues and value-sensitive design into engineering specifications
4. Develop measures of merit based on finalized engineering specifications
5. Create a decision matrix template and assign responsibilities for Part 3
6. Fix .m file from ICE13

Discussion:
In today's meeting, the team discussed the requirements for Final Project Part 3 and developed a plan for completing the deliverable. Since feedback from Part 2 has not been released, the team agreed that it will be reviewed immediately upon receipt, and any necessary revisions will be incorporated into the customer requirements and engineering specifications.

The team revisited the bus tracking screen project and discussed how value-sensitive design and virtues introduced in recent lectures could be incorporated into the design process. Accessibility, reliability, inclusivity, transparency, and sustainability were identified as important virtues that should be considered when finalizing customer requirements and engineering specifications. The team discussed possible ways to quantify these values through measurable design criteria.

The team also began planning the development of measures of merit. Team members reviewed the engineering specifications created in Part 2 and discussed how each specification could be translated into a measurable state variable. The team identified the need to determine whether each measure of merit should be maximized or minimized and establish appropriate target values.

Finally, the team discussed the decision matrix requirements for Part 3. The team reviewed the concept evaluation process presented in Lecture 13 and outlined the structure of the decision making template, including comparison criteria, weighting factors, concept alternatives, absolute scores, weighted scores, and customer satisfaction calculations. A mock decision matrix will also be created to verify that the template functions correctly before concept generation begins.

Decisions Made:
1. Review Part 2 feedback immediately when it becomes available
2. Begin finalizing engineering specifications and customer requirements in preparation for revisions
3. Incorporate virtues and value-sensitive design principles into the final engineering specifications

4. Develop measures of merit for each engineering specification 5. Create a decision matrix template for future concept evaluation 6. Generate a mock decision matrix to verify calculations and spreadsheet functionality		
Action Items	Person Responsible	Deadline
Review Part 2 feedback and identify required revisions once released	All	Upon release of feedback
Finalize list of virtues and determine how they will be quantified in engineering specifications	Tyler	June 10
Develop measures of merit for each engineering specification	Ava	June 10
Create decision matrix template spreadsheet with scoring and weighting calculations	Sarah	June 10
Generate mock decision matrix using conceptual data to verify functionality with .m file	Odon	June 11
Team member: Ava Towery	Date for next meeting:	
Team member: Sarah Senn	Wed June 10, 2026	
Team member: Tyler Asmussen		
Team member: Odon Ineza		

The Mechanical Design Process

Team Meeting Minutes

Design Organization: Group 1	**Date: Wednesday June 10, 2026**

Agenda:

1. Review the changes in the assignment as outlined in the new rubric and Dr. Goehler's email.
2. Review each team member's work related to the previous Part 3 deliverables divided among the team.
3. Generate new virtues that will govern the newly generated engineering specifications.
4. Generate new engineering specifications informed by those virtues.
5. Create a value-sensitive design analysis for the newly generated engineering specifications.
6. Understand what the measures of merit should be based on the finalized engineering specifications.
7. Make a submission checklist and prepare points of feedback.

Discussion:
The team began by reviewing the updated rubric and Dr. Goehler's email outlining the changes to Final Project Part 3. New requirements were identified and mapped against the work already completed, allowing the team to determine which portions needed to be revised, added, or replaced before submission.

Each team member then presented a status update on the portion of the Part 3 work they had been assigned at the previous meeting. Sarah presented the decision matrix template, Ava presented the measures of merit, Tyler presented the virtue quantification analysis, and Odon demonstrated the mock decision matrix built using the calculateMoM.m function from ICE13. The team reviewed each deliverable against the updated rubric requirements and identified specific gaps or corrections needed.

Following the individual reviews, the team focused on generating new virtues to govern the engineering specifications. Starting from a brainstormed list of five candidate virtues — accessibility, honesty, sustainability, reliability, and inclusivity — the team discussed the relevance of each to the Dublin bus display system. After careful deliberation, the team narrowed the list to two primary virtues that were judged most directly applicable and most amenable to quantification in the context of this project.

With the virtues established, the team moved on to generating a base set of two new engineering specifications directly derived from those virtues. Each new specification was discussed in terms of its state variable, measurement unit, direction of improvement, and target value to ensure it was ready for use in the measures of merit and decision matrix.

The team then conducted a value-sensitive design (VSD) analysis for the newly generated engineering specifications, discussing how the chosen virtues are embedded in the design criteria and what tradeoffs or stakeholder impacts may arise. The analysis was reviewed and approved by all team members before moving on.

The full list of measures of merit was then assembled and reviewed. For each MOM, the team confirmed the state variable name, the appropriate inputs to calculateMoM() (min_val, max_val, and direction), and the optimal value that returns a score of 10. Each item was discussed and approved by the team.

Finally, the team reviewed the decision matrix template and mock decision matrix to confirm alignment with the new rubric. The polished decision matrix was agreed upon. The team also built a submission checklist to track all required deliverables and identified specific points of feedback to seek from the teaching team before confirming the deliverable is ready to submit.

Decisions Made:

1. New points of work were identified and assigned based on the updated rubric and Dr. Goehler's email.
2. A base number of two new engineering specifications was established, each grounded in one of the two selected virtues.

3. From a brainstormed list of five candidate virtues, the team narrowed the selection to the two most important and quantifiable virtues for the xRank project.
4. A value-sensitive design analysis for the newly generated engineering specifications was reviewed and agreed upon by the full team.
5. The complete list of measures of merit, including all inputs to calculateMoM() (state variable, min_val, max_val, direction), was generated and approved by all team members.
6. The final polished decision matrix template and mock decision matrix were reviewed and agreed upon.
7. The submission checklist was completed and reviewed by the team.
8. The team agreed on specific points of feedback to seek from the instructor before confirming readiness of the deliverable.

Action Items	Person Responsible	Deadline
Review Part 2 feedback upon release and incorporate revisions into customer requirements and engineering specifications.	All	June 10
Finalize virtue list (2+ virtues) and document how each was quantified in the engineering specifications.	Ava	June 10
Update and finalize measures of merit table, including all calculateMoM() inputs.	Tyler	June 10
Finalize decision matrix template spreadsheet with all formula-driven scoring, weighting.	Sarah	June 10
Compile all deliverables against submission checklist and prepare draft for team review.	Odon	June 11

Team member:	Ava Towery	Date for next meeting: June 11, 2026
Team member:	Sarah Senn	
Team member:	Tyler Asmussen	
Team member:	Odon Ineza	

Team Meeting Minutes

Design Organization: Group 1	**Date: Monday June 15, 2026**

Agenda:

1. Discuss Project Part 2
2. Read over Project Part 4
3. Brainstorm multiple initial ideas for potential physical/digital functions that could meet customer requirements.

Discussion:

From the feedback in Part 2, we need to add citations for our customer requirements and engineering specifications. There was discussion about whether to add these in a paragraph or another column in the table. It was decided to add these to each table so it would be clear where each requirement and specification came from.

Next, the team brainstormed ideas for function for Part 4. Time was spent breaking down the assignment to figure out where to start. The goal is to come up with at least 6 ideas of a physical/digital function and concepts for design that go along with them. For today, we need to think about the specifications but don't connect them to the specifications yet.

Original subsystem ideas:
Physical subsystem: dealing with safety, and weather/vandalism
Digital subsystems: real-time bus tracking and processing, displaying different languages, displaying accurate data, sustainability, accessibility, displaying accurate data

Once the subsystems were created, then functions that enforce those subsystems had to be generated.

Decisions Made:

1. Add citations for references to both customer requirements and engineering specifications into the tables.

Action Items	Person Responsible	Deadline
Add citations to requirements and specifications	Odon	6/15
Brainstorm initial ideas	All	6/16

Team member: Ava Towery	Date for next meeting:	
Team member: Sarah Senn	June 16, 2026	
Team member: Tyler Asmussen		
Team member: Odon Ineza		

Team Meeting Minutes

Design Organization: Group 1	**Date: Wednesday June 17, 2026**

Agenda:

- Fix errors from part 3
- Generate functional concepts for subsystems
- Pick one subsection to focus on
- Create concepts and develop sketches for each of the functions of the subsection

Discussion:

During today's meeting, the team agreed to focus on two main tasks, brainstorming for Part 4 of the assignment and correcting issues identified in Part 3. The group recognized that several revisions to Part 3 are essential before continuing, including placing references in the correct order, reorganizing sections to improve the document's structure, switching some of the measures of merit to binary/pass/fail table(constant must have/state variables), and correcting the meeting minutes dates and confirming them. Addressing these issues now will make future revisions more manageable and prevent more extensive changes later as additional content is added.

To divide the workload efficiently, one team member will temporarily focus on implementing these structural corrections while the remaining members collaborate on the brainstorming activities. After reviewing and clarifying the objectives of Part 4, the team concluded that the brainstorming should concentrate on selecting and thoroughly exploring one subsystem in greater detail.

Decisions Made:

- The team will focus on the subsystem about how to incorporate the multiple languages onto the RTVI display/at bus stops effectively
- First fix errors from comments on Part 3, then focus on requirements for part 4

Action Items	Person Responsible	Deadline

Fix reference errors from part 3	Sarah	June 18
Create sketches	Tyler	June 18
Create descriptions about functions of each sketch	Ava	June 18
Add virtues component for sketches/subsystem and respective functions	Odon	June 19
Generate correct decision matrix and constant variable chart	Tyler	June 19

Team member: Ava Towery	Date for next meeting: June 18, 2026
Team member: Sarah Senn	
Team member: Tyler Asmussen	
Team member: Odon Ineza	

The Mechanical Design Process

Team Meeting Minutes

Design Organization: Group 1	**Date: Thursday June 18, 2026**

Agenda:

- Continue revisions from Part 3 feedback
- Refine functional concepts and sketches for the multilingual display subsystem and its respective functions
- Discuss virtues incorporated into brainstorming and function generation
- Begin developing state variables and quantitative models for function evaluation
- Discuss concept selection and decision matrix criteria - create new table for constant engineering specifications

Discussion:

During todays meeting, the team continued working on Final Project Part 4 while also addressing revisions identified from feedback on Part 3. The group reviewed the remaining changes that need to be made to the report, including separating pass/fail engineering requirements from measures of merit, strengthening the engineering ethics discussion within the VSD section, and reorganizing sections of the report to improve the overall flow and readability.

The team then continued development of the selected subsystem concerning how multiple languages should be displayed on the RTVI system. Previously generated functional concepts were reviewed and refined, including the button interactive system to change languages, multilingual audio announcements, symbols and pictograms, automatic language cycling over time, and simultaneous multilingual display. The team discussed the strengths and weaknesses of each concept, and is in the making of each sketch for each function in the report via GoodNotes on ipad.

The virtues of inclusivity, accessibility, reliability, and community awareness were revisited and conceptualized with each function in the subsystem to determine how they influenced the generation of the functions and how they would be incorporated into the evaluation process.

The group additionally began identifying state variables and model that can be used to quantitatively evaluate the subsystem functions. Potential variables related to multilingual support, accessibility, implementation, information accessibility, determination of use, and connecting them to the engineering specifications and measures of merit already developed, maybe having to create new measures of merit/engineering specifications to satisfy the models.

Finally, the team discussed the concept selection process and the development of the decision matrix. Initial discussion focused on which customer requirements and engineering specifications should be used as comparison criteria and how weighting factors can be justified using the rankings established in previous deliverables.

Decisions Made:

- Continue implementing revisions from Part 3 feedback before submission
- Refine and finalize the 5 functions concepts for the multilingual display subsystem; create and finalize sketches
- Use inclusivity, accessibility, reliability, and community awareness as primary virtues discussed in the brainstorming section
- Begin developing quantitative models and state variables tied to engineering specifications and measures of merit
- Use customer requirement rankings and engineering specification rankings to justify decision matrix weights

Action Items	Person Responsible	Deadline
Finalize and label sketches for all functions	Tyler	June 19
Complete descriptions for each function	Ava	June 19
Format and rewrite full LaTeX file for readability - Fix Part 3 comments for Part 4	Sarah	June 19
Create and develop models for subsystem concepts	Odon	June 19

Team member:	Ava Towery	Date for next meeting:
Team member:	Sarah Senn	June 19, 2026
Team member:	Tyler Asmussen	
Team member:	Odon Ineza	